

# CHAPTER 3

## Unit 2

# Heat Transfer Through Extended Surfaces (Fins)

### Syllabus :

Extended Surfaces.

## 3.1 Introduction to Extended Surfaces

There are large numbers of engineering equipments where the unutilized heat energy is required to be dissipated / discarded to the atmosphere. If this heat is not dissipated, the system will fail due to overheating.

Examples of engineering systems needing the heat dissipation are the cooling of internal combustion engines, heat removal from nuclear reactors, cooling of electronic components / integrated circuits, transformers, motors, compressors, refrigerators etc.

In most of these cases the heat transfer takes place by conduction and convection. Heat is conducted in solid material upto its surface and finally the heat is rejected by convection from its surface to the surroundings.

*Thus the aim is to increase the heat transfer rates from the surface so that the temperature of solid surface is maintained within desired limits to avoid failure of the system.*

## 3.2 Extended Surfaces-Fins

The rate of heat transfer from solid surface to fluids / atmosphere is given by the equation :

$$Q = h \cdot A (T - T_{\infty}) \text{ where,}$$

$h$  = convective heat transfer coefficient

$A$  = Surface area

$(T - T_{\infty})$  = Temperature difference between the solid surface and the fluid / ambient air.

In above heat transfer equation, the value of  $h$  is almost constant ( $5$  to  $12 \text{ W / m}^2 \text{ K}$ ) whenever the heat is convected to atmosphere and the temperature difference  $(T - T_{\infty})$  can not be controlled.

Therefore, the only way is to increase the heat transfer rate,  $Q$  is by increasing the surface area  $A$ . This surface area of solid is increased by providing extended surfaces called fins.

The heat is transferred from liquid to gas or gas to gas or liquid to liquid through a solid conducting surfaces. The convective heat transfer coefficient for gas ( $h_g$ ) is usually much smaller than the heat transfer coefficient of liquids ( $h_l$ ). Obviously the convective heat transfer between solid and gas is much smaller than the convective heat transfer between solid and liquids. **In order to compensate for low convective heat transfer coefficient in case of gases compared to liquids, the surface area on the gas side can be extended by providing fins. However, fins are provided on lower  $h$  side.**

### 3.2.1 Fins on Lower 'h' Side

Consider Fig. 3.2.1, where a plate has different fluids on its two sides having higher ' $h$ ' as  $h_H$  on left and lower ' $h$ ' as  $h_L$  on right side. Assume that heat flow is taking place from  $T_H$  to  $T_L$ ;  $T_H$  and  $T_L$  being higher and lower temperatures of fluids.

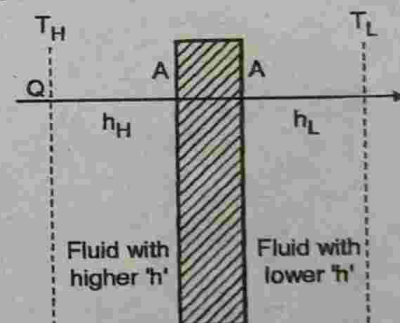


Fig. 3.2.1

As per concept, restricting factor for heat flow  $Q$  shall be the side, where more resistance is acting to heat flow.

Since convective resistance is given as  $\frac{1}{hA}$ , resistance on right side  $R_R = \frac{1}{h_L A}$  will be higher than convective resistance acting on left face  $R_L = \frac{1}{h_H A}$

Therefore, to enhance heat transfer rate by fins, fins should be used on lower 'h' side as  $Q = hA\Delta T$ , here area A can be increased by using fins.

If fins are fitted on higher 'h' side, it shall be ineffective arrangement.

### 3.3 Types of Fins and Applications

The fins are normally thin strips of high conducting materials such as copper, aluminium, steel, brass etc.

Fins can be of different shapes and sizes depending upon its application as shown in Fig. 3.3.1.

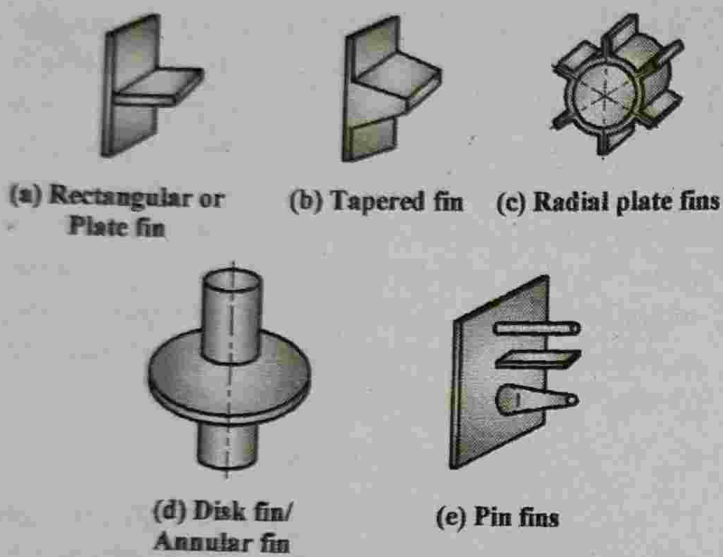


Fig. 3.3.1 : Types of fins

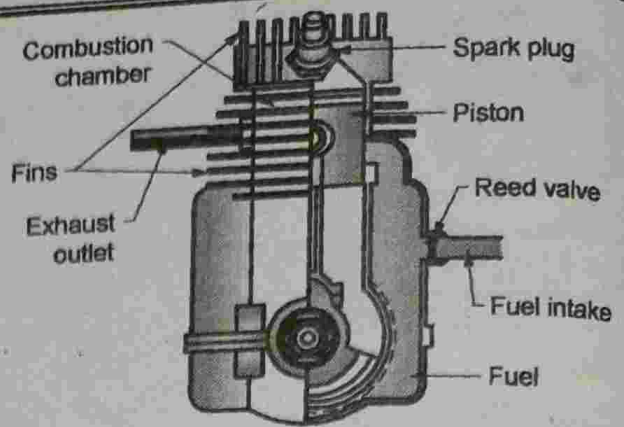
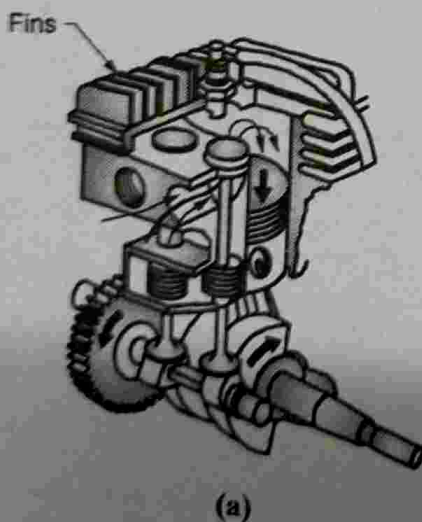


Fig. 3.3.2 : Plate fins in I.C. Engines

The fins which are attached to the plane walls are called **straight fins**. These fins may be of constant cross-sectional area like **rectangular plate fins** or it may be of varying cross-sectional area like **tapered fins**, **radial plate fins** etc. The use of plate fins in I.C. engine is depicted in Fig. 3.3.2.

An **annular fin** is a fin attached circumferentially to a cylinder. The **pin fins** or **spine** is an extended surface of circular cross-section. Usually the diameter of pin fins is much less than its length. The pin fins may also be uniform or varying cross-section according to the requirement and application.

#### Applications

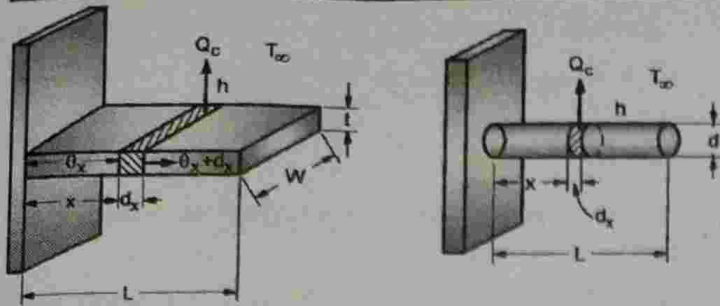
As discussed above, usually the fins are provided on the surface where convective heat transfer coefficient is low as in case of gases e.g. extended surfaces are used in radiators, cylinders of compressors and engines, transformers, motors, various types of heat exchangers etc.

#### Selection of fins

The geometry of fins are selected based on the cost, efficiency, performance, weight, availability of space, pressure drop in the heat transfer fluid etc.

### 3.4 Analysis of Fins of Uniform Cross-Sectional Area (Rectangular Plate Fin / Pin Fin)

Consider a thin rectangular fin of uniform cross-sectional area as shown in Fig. 3.4.1(a) or a pin fin of uniform cross-sectional area shown in Fig. 3.4.1(b). In each case the fin is attached to the base surface at temperature  $T_0$ .



Cross-sectional area,  $A_c = w \cdot t$

$$A_c = \frac{\pi}{4} d^2$$

Perimeter,  $P = 2(w + t) \approx 2w$  if  $(t \ll w)$

$$P = \pi \cdot d$$

Length,  $L$

Length =  $L$

(a) Thin rectangular plate fin

(b) Pin fin

**Fig. 3.4.1 : Analysis of fins of uniform cross-sectional area**

Following **assumptions** are made in analysis of pin fins :

1. Heat conduction is along the x-axis under steady state.
2. Convective heat transfer coefficient over the fin is uniform over the entire surface.
3. There is no heat generation within the fin.
4. Material is isotropic i.e.  $k$  is constant.
5. Base temperature is maintained constant at  $T_0$ .

Let  $A_c$  be cross-sectional area and  $P$  be the perimeter of the fin,  $T_\infty$  the atmospheric temperature and  $L$  as length of the fin.

The heat is conducted from the base surface through the fin and to atmospheric surrounding fluid from fin surface by convection simultaneously.

Consider heat flow to an elemental section of thickness  $dx$  at a distance  $x$  from the base temperature,  $T_0$ .

Heat entering into elementary section,

$$Q_x = -k \cdot A \cdot \frac{dT}{dx} \quad \dots(i)$$

Heat conducted out from element at  $(x + dx)$ ,

$$Q_{x+dx} = Q_x + \frac{d}{dx} (Q_x) \cdot dx \quad \dots(ii)$$

Heat convected out from surface of element of thickness,  $dx$ ,

$$Q_c = h \cdot P \cdot dx \cdot (T - T_\infty) \quad \dots(iii)$$

where,  $T$  is the temperature at section  $x$ ,  $(P \cdot dx)$  is the area for convection.

Writing energy balance equation for element  $dx$  :

Heat conducted into element = Heat conducted out + Heat convected out.

$$Q_x = Q_{x+dx} + Q_c$$

$$Q_x = Q_x + \frac{d}{dx} (Q_x) \cdot dx + h \cdot P \cdot dx \cdot (T - T_\infty)$$

$$-\frac{d}{dx} (Q_x) \cdot dx = h \cdot P \cdot dx \cdot (T - T_\infty)$$

$$-\frac{d}{dx} \left( -k \cdot A \cdot \frac{dT}{dx} \right) \cdot dx = h \cdot P \cdot dx \cdot (T - T_\infty)$$

$$k \cdot A \cdot \frac{d^2 T}{dx^2} = h \cdot P \cdot (T - T_\infty)$$

$$\frac{d^2 T}{dx^2} = \frac{h \cdot P}{k \cdot A} (T - T_\infty) \quad \dots(iv)$$

$$\text{Let, } (T - T_\infty) = \theta, \text{ the excess temperature.} \quad \dots(v)$$

On differentiating above equation twice

$$\frac{dT}{dx} = \frac{d\theta}{dx}$$

$$\text{and } \frac{d^2 T}{dx^2} = \frac{d^2 \theta}{dx^2} \quad \dots(vi)$$

$$\text{Let, } \frac{h \cdot P}{k \cdot A} = m^2 \text{ (for convenience)} \quad \dots(3.4.1)$$

Using Equations (vi) and (3.4.1), the above Equation (iv) takes the form,

$$\frac{d^2 \theta}{dx^2} - m^2 \cdot \theta = 0 \quad \dots(3.4.2)$$

Equation (3.4.2) is the second order differential equation. It represents the temperature distribution along the fin.

The general solution to the above second order differential Equation is,

$$\theta = C_1 \cdot e^{mx} + C_2 \cdot e^{-mx} \quad \dots(3.4.3)$$

And the slope,

$$\frac{d\theta}{dx} = m \cdot C_1 \cdot e^{mx} - m \cdot C_2 \cdot e^{-mx} \quad \dots(3.4.4)$$

In order to solve the Equation (3.4.4), we need to determine the constants  $C_1$  and  $C_2$  for which we need two boundary conditions. We shall obtain the solution of the above equation by considering the following three cases :

**Case I :** The fin is infinitely long and the temperature at the end of fin is essentially equal to surrounding fluid temperature.

**Case II :** Adequately long fin with insulated end tip.

**Case III :** The short fin which loses heat by convection from fin cross sectional area at for end.

### 3.4.1 Case I : Analysis of Infinitely Long Fin

An infinitely long fin is shown in Fig. 3.4.2. In such a case the temperature at the end of pin approaches to surrounding fluid temperature,  $T_\infty$  as shown in Fig. 3.4.2. Such a condition can be practically met by a 0.5 mm diameter rod which is about 8 cm long.

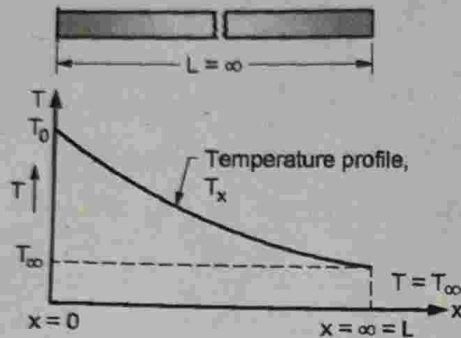


Fig. 3.4.2 : Infinitely long fin

Boundary conditions (B.C.) are : ( $\theta = T - T_\infty$ )

1. At  $x = 0$ ,  $T = T_0$   
i.e.  $\theta = T_0 - T_\infty$
2. At  $x = L \rightarrow \infty$ ,  $T = T_\infty$   
i.e.  $\theta = 0$

$$\theta = C_1 \cdot e^{mx} + C_2 \cdot e^{-mx} \quad \dots(\text{from Equation 3.4.3})$$

On applying boundary condition (1) in Equation (3.4.3) we get,

$$\theta_0 = C_1 + C_2 \quad \dots(\text{i})$$

On applying boundary condition (2) in Equation (3.4.3) we get,

$$0 = C_1 \cdot e^{m \cdot \infty} + C_2 \cdot e^{-m \cdot \infty}$$

$$0 = C_1 \times (\infty) + 0$$

This equality can hold good if

$$C_1 = 0 \quad \dots(\text{ii})$$

From Equations (i) and (ii),  $\theta_0 = 0 + C_2$

$$\text{i.e. } C_2 = \theta_0 = T_0 - T_\infty \quad \dots(\text{iii})$$

On substituting the values of  $C_1$  and  $C_2$  in Equation (3.4.3) we get :

$$\theta = 0 + (T_0 - T_\infty) e^{-mx} \quad \dots(\text{3.4.5})$$

$$\frac{d\theta}{dx} = -(T_0 - T_\infty) m e^{-mx} \quad \dots(\text{iv})$$

Rewriting equation (2.8.5)

$$T - T_\infty = (T_0 - T_\infty) e^{-mx} \quad \dots(\text{3.4.6})$$

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{-mx} \quad \text{or} \quad \frac{\theta}{\theta_0} = e^{-mx} \quad \dots(\text{3.4.7})$$

Above equation represents the temperature distribution in an infinitely long fin.

The heat flow through the fin can either be calculated by heat conduction from base or by considering the heat transfer by convection from the entire fin.

**Heat transfer by conduction at base**

$$\begin{aligned} Q &= -k \cdot A \cdot \left( \frac{d\theta}{dx} \right)_{x=0} \\ &= -k \cdot A \cdot [-(T_0 - T_\infty) m \cdot e^{-mx}]_{x=0} \\ Q &= +k \cdot A \cdot m \cdot (T_0 - T_\infty) \\ &= k \cdot A \times \sqrt{\frac{h \cdot P}{k \cdot A}} (T_0 - T_\infty) \\ &= \sqrt{k \cdot A \cdot h \cdot P} (T_0 - T_\infty) \quad \dots(\text{3.4.8}) \end{aligned}$$

Heat transfer by convection,

$$\begin{aligned} Q_c &= \int_0^\infty h \cdot P \cdot dx \cdot (T - T_\infty) \\ &= \int_0^\infty h \cdot P \cdot (T_0 - T_\infty) \cdot e^{-mx} \cdot dx \\ &= h \cdot P \cdot (T_0 - T_\infty) \left[ -\frac{1}{m} \cdot e^{-mx} \right]_0^\infty \\ &= h \cdot P \cdot (T_0 - T_\infty) \left[ -\frac{1}{m} \cdot (e^{-m \cdot \infty} - e^{-m \cdot 0}) \right] \\ &= h \cdot P \cdot (T_0 - T_\infty) \left( -\frac{1}{m} \right) (0 - 1) \\ &= \frac{h \cdot P (T_0 - T_\infty)}{m} = \frac{h \cdot P (T_0 - T_\infty)}{\sqrt{\frac{h P}{k A}}} \\ &= \sqrt{k \cdot A \cdot h \cdot P} (T_0 - T_\infty) \quad \dots(\text{v}) \end{aligned}$$

Thus the above Equation (iv) is same as Equation (3.4.8).

$$\text{i.e. } Q_{\text{at } x=0} = Q_c$$

### 3.4.2 Case II : Adequately Long Fin with Insulated End Tip

Refer Fig. 3.4.3.

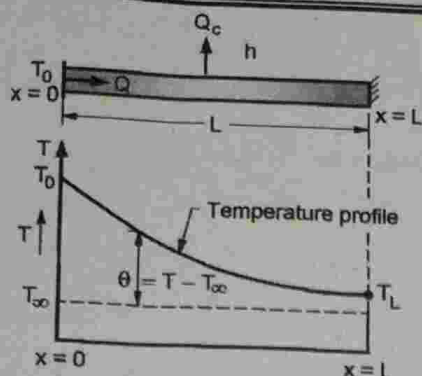


Fig. 3.4.3 : Long fin with insulated end tip

Practically the heat loss from the long and thin fin tip is negligible, thus the end of the tip can be considered as insulated.

**Boundary conditions are :**

1. At  $x = 0$ ,  $T = T_0$  and  $\theta = T_0 - T_\infty = \theta_0$

2. At  $x = L$ ,  $Q = 0$  i.e.  $\frac{d\theta}{dx} = 0$

$$\theta = C_1 e^{mx} + C_2 e^{-mx} \quad \dots(i)$$

On applying first boundary condition in Equation (i) we get,

$$\theta_0 = C_1 + C_2 \quad \dots(ii)$$

Differentiating Equation (i),

$$\frac{d\theta}{dx} = m C_1 e^{mx} - m C_2 e^{-mx} \quad \dots(iii)$$

On applying boundary condition (2) we get,

$$0 = m C_1 e^{mL} - m C_2 e^{-mL}$$

$$\therefore C_2 = C_1 \frac{e^{mL}}{e^{-mL}} \quad \dots(iv)$$

Combining Equation (ii) and (iv),

$$\theta_0 = C_1 + C_1 \cdot \frac{e^{mL}}{e^{-mL}} = C_1 \left( \frac{e^{-mL} + e^{mL}}{e^{-mL}} \right)$$

$$C_1 = \theta_0 \left( \frac{e^{-mL}}{e^{-mL} + e^{mL}} \right) \quad \dots(v)$$

Using  $C_1$  in Equation (iv),

$$\begin{aligned} C_2 &= \theta_0 \left( \frac{e^{-mL}}{e^{-mL} + e^{mL}} \right) \times \frac{e^{mL}}{e^{-mL}} \\ &= \theta_0 \left( \frac{e^{mL}}{e^{-mL} + e^{mL}} \right) \quad \dots(vi) \end{aligned}$$

On substituting the values of  $C_1$  and  $C_2$  from Equations (v) and (vi) in Equation (i) we get,

$$\theta = \theta_0 \left[ \frac{e^{-mL} \cdot e^{mx}}{e^{-mL} + e^{mL}} \right] + \theta_0 \left[ \frac{e^{mL} \cdot e^{-mx}}{e^{-mL} + e^{mL}} \right]$$

$$\frac{\theta}{\theta_0} = \frac{e^{-mL} \cdot e^{mx} + e^{mL} \cdot e^{-mx}}{e^{-mL} + e^{mL}}$$

$$\frac{\theta}{\theta_0} = \frac{e^{-m(L-x)} + e^{m(L-x)}}{e^{-mL} + e^{mL}}$$

$$= \frac{\cosh m(L-x)}{\cosh mL} \quad \dots(3.4.9)$$

$$\left[ \text{Note that : } \cosh \alpha = \frac{e^\alpha + e^{-\alpha}}{2} \right]$$

But  $\theta = T - T_\infty$

and  $\theta_0 = T_0 - T_\infty$ , Therefore,

$$\frac{\theta}{\theta_0} = \frac{T - T_\infty}{T_0 - T_\infty} = \frac{\cosh m(L-x)}{\cosh mL} \quad \dots(3.4.10)$$

Temperature at the tip,  $T_L$  i.e. at  $x = L$  will be :

$$\frac{T_L - T_\infty}{T_0 - T_\infty} = \frac{1}{\cosh mL} \quad \dots(3.4.11)$$

Heat flow rate from fin  $Q$  : Heat conducted to fin at the base shall be the heat flow rate from fin. Hence,

$$Q = -k A \left( \frac{d\theta}{dx} \right)_{x=0} \quad \dots(vii)$$

On differentiating Equation (2.8.9),

$$\frac{d\theta}{dx} = \theta_0 \cdot \left[ \frac{-m \sinh m(L-x)}{\cosh mL} \right] \quad \dots(viii)$$

From Equations (vii) and (viii),

$$Q = -k \cdot A \cdot \theta_0 \left[ \frac{-m \sinh m(L-x)}{\cosh mL} \right]_{x=0}$$

$$= -k \cdot A \cdot \theta_0 \cdot m \left[ \frac{-\sinh m \cdot L}{\cosh mL} \right]$$

$$\begin{aligned} Q &= k \cdot A \cdot \theta_0 \cdot m \tanh mL \\ &= \sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \tanh mL \quad \dots(3.4.12) \end{aligned}$$

$$\therefore \left( m = \sqrt{\frac{h \cdot P}{k \cdot A}} \right)$$

In case of infinitely long fins,  $\tanh mL = 1$

The Equation (3.4.12) reduces to :

$$\begin{aligned} Q &= \sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \\ &= \sqrt{h \cdot P \cdot k \cdot A} (T_0 - T_\infty) \end{aligned}$$

...[Same as Equation 3.4.8]

The variation of  $Q$  Vs  $mL$  and variation of  $\frac{\theta}{\theta_0}$  can be plotted by considering Equations (3.4.12) and (3.4.10) respectively as shown in Fig. 3.4.4. According to these equations,  $Q \propto \tanh mL$  and  $\frac{\theta}{\theta_0} \propto \frac{1}{\cosh mL}$ .

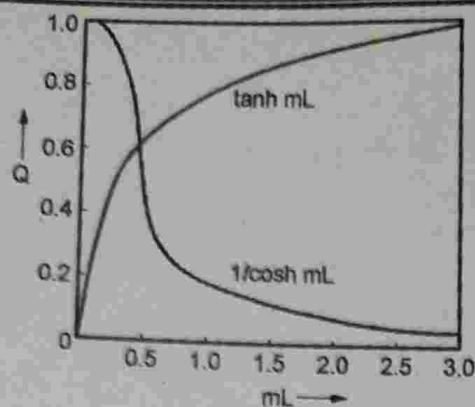


Fig. 3.4.4 : Variation of  $Q$  and  $\frac{\theta}{\theta_0}$  along the length of thin adequately long fin

As  $mL$  or  $L$  increases, the value of  $\tanh mL$  first increases rapidly and then the rate slowly decreases and finally it becomes asymptotic at  $mL \approx 3$ .

Also value of  $\theta$  at  $mL \approx 3$  approaches zero. It implies that increasing the value of  $mL$  beyond 3 will not substantially increase the heat transfer rate. Hence, fin with  $mL \approx 3$  is called *adequately long fin* or *fin with insulated tip* or *adiabatic tip*.

### 3.4.3 Case III : Analysis of Short Fin

The boundary conditions are

1. At  $x = 0$ ,  $T = T_0$  and  $\theta = \theta_0$
2. At  $x = L$ ,  $Q = Q_c$

This is referred as the condition for a short or finite fin where the rate of heat conducted to the end is convected away to the surroundings according to boundary condition (2) specified above. Hence,  $Q = Q_c$  i.e.

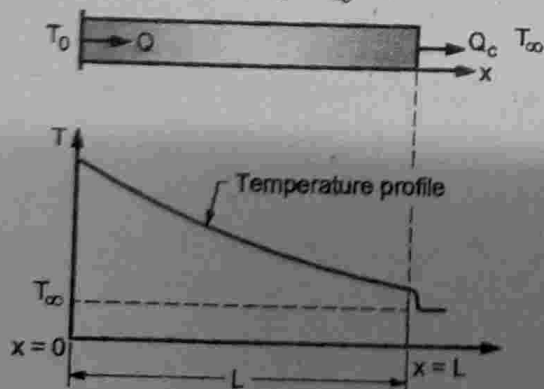


Fig. 3.4.5 : Heat transfer by convection at the tip

$$-k \cdot A \cdot \left( \frac{d\theta}{dx} \right)_{x=L} = [h \cdot A \cdot \theta]_{x=L} \quad \dots(i)$$

(where  $\theta = T - T_\infty$ )

$$\text{Since, } \theta = C_1 \cdot e^{mx} + C_2 \cdot e^{-mx} \quad \dots(ii)$$

$$\therefore \frac{d\theta}{dx} = C_1 \cdot m \cdot e^{mx} - C_2 \cdot m \cdot e^{-mx} \quad \dots(iii)$$

Combining Equations (i) to (iii),

$$-k \cdot A \cdot [m \cdot C_1 e^{mx} - m \cdot C_2 e^{-mx}]_{x=L} = h \cdot A \cdot [C_1 e^{mx} + C_2 e^{-mx}]_{x=L}$$

$$-k \cdot A \cdot m [C_1 e^{mL} - C_2 e^{-mL}] = h \cdot A [C_1 e^{mL} + C_2 e^{-mL}]$$

$$C_1 \cdot e^{mL} (k \cdot m + h) = C_2 \cdot e^{-mL} (k \cdot m - h) \quad \dots(iv)$$

Using boundary condition (1) in Equation (ii) as  $x = 0$ ,  $\theta = \theta_0$  we get,

$$\theta_0 = C_1 + C_2 \quad \dots(v)$$

On solving Equations (iv) and (v)

$$C_1 = \frac{\theta_0 \left( 1 + \frac{h}{mk} \right)}{(e^{mL} + e^{-mL}) + \left( \frac{h}{mk} \right) (e^{mL} - e^{-mL})} \quad \dots(3.4.13)$$

$$C_2 = \theta_0 - \frac{\theta_0 \left( 1 + \frac{h}{mk} \right)}{(e^{mL} + e^{-mL}) + \left( \frac{h}{mk} \right) (e^{mL} - e^{-mL})} \quad \dots(3.4.14)$$

On substituting the values of  $C_1$  and  $C_2$  in Equation (ii) we get,

$$\frac{\theta}{\theta_0} = \frac{\cosh m(L-x) + \frac{h}{mk} \cdot \sinh m(L-x)}{\cosh mL + \frac{h}{mk} \cdot \sinh mL} \quad \dots(3.4.15)$$

At  $x = L$ , let  $\theta = \theta_L$

$$\frac{\theta_L}{\theta_0} = \frac{1}{\cosh mL + \frac{h}{mk} \cdot \sinh mL} \quad \dots(3.4.16)$$

Heat flow rate from fin,  $Q$

$$Q = -k \cdot A \cdot \left( \frac{d\theta}{dx} \right)_{x=0}$$

On substituting the value of  $\frac{d\theta}{dx}$  at  $x = 0$  we get,

$$Q = k \cdot A \cdot m \cdot \theta_0 \cdot \frac{\sinh(mL) + \frac{h}{mk} \cosh(mL)}{\cosh(mL) + \frac{h}{mk} \sinh(mL)}$$

Substituting,  $m = \sqrt{\frac{Ph}{kA}}$  and dividing the numerator and denominator by  $\cosh(mL)$  in the above equation we get:

$$Q = \sqrt{h P k A} \cdot \theta_0 \cdot \left[ \frac{\tanh(mL) + \frac{h}{mk}}{1 + \frac{h}{mk} \cdot \tanh(mL)} \right] \quad \dots(3.4.17)$$

### 3.5 Effectiveness of Fins

The aim of providing the fins is to increase the heat transfer rate from a surface by increasing the surface area. However, the fin itself blocks the base area of the surface and acts as a resistance to heat conduction. Therefore, the heat transfer from the base surface decreases. Thus, one is not sure whether the heat transfer rates will increase or not by providing the fins on the base surface. It is assessed by a term called *fin effectiveness* ( $E$ ).

#### Definition of Fin effectiveness

It is defined as the ratio of actual heat transfer rate from fin surface to the heat transfer rate from the area blocked by fin (i.e. as if there was no fin on base surface).

Effectiveness,

$$E = \frac{\text{Actual heat transfer from fin surface, } Q}{\text{Rate of heat transfer without fins}} \quad \dots(3.5.1)$$

Naturally the use of fins is justified only when effectiveness of fin is greater than 1. Practically the use of fins increases the cost of the system. Thus the provisioning of the fins is only justified by taking into consideration the cost element when effectiveness,  $E$  is more than 2.

In case of insulated tip,

$$\begin{aligned} E &= \frac{Q}{h \cdot A \cdot \theta_0} \\ &= \frac{\theta_0 \sqrt{P h k A} \cdot \tanh mL}{h \cdot A \cdot \theta_0} = \frac{\tanh mL}{\sqrt{\frac{h A}{P k}}} \end{aligned}$$

When  $mL$  is large i.e.  $mL \geq 3$ ,  $\tanh(mL)$  tends to become 1. Hence, for infinite long fin,

$$E = \sqrt{\frac{k \cdot P}{h \cdot A}} \quad \dots(3.5.2)$$

#### Important Conclusions

- Effectiveness increases with increase in  $k$ . Hence materials used for fins must have high thermal conductivity e.g. materials like copper, aluminium etc. However, the weight is also taken into consideration while choosing the materials.
- Effectiveness increases with increase in  $(P/A)$ , for this reason thin fins are preferred.

- Effectiveness increases if the convective heat transfer coefficient ' $h$ ' is low. For this reason the fins are provided on the gas side having natural convection.

### 3.6 Fin Efficiency ( $\eta$ )

#### Definition

Fin efficiency is defined as the ratio of actual heat transfer rate to the maximum possible heat transfer rate from the same fin.

Heat transfer rate,  $Q$  from the fin shall be maximum when the fin material has infinite thermal conductivity ( $k$ ) so that the temperature of fin all along the length can be assumed to be same as that at the base of the fin.

[ Note :  $Q \propto (T_0 - T_\infty)$  from base ]

$$\therefore \eta = \frac{\text{Actual heat transfer rate from fin, } Q}{\text{Maximum possible heat transfer rate, } Q_{\max}} \quad \dots(3.6.1)$$

$$\eta = \frac{Q}{h \cdot A_f \cdot \theta_0} = \frac{Q}{h \cdot P \cdot L \cdot \theta_0} \quad \dots(3.6.2)$$

where  $A_f$  is the total fin surface area i.e.  $A_f = P \cdot L$ .

Actual heat transfer,  $Q = \sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \tanh mL$

For long fin with insulated tip.

$$\begin{aligned} \therefore \eta &= \frac{\theta_0 \cdot \sqrt{h \cdot P \cdot k \cdot A} \tanh mL}{h \cdot P \cdot L \cdot \theta_0} \\ &= \left( \frac{k \cdot A}{h \cdot P} \right)^{1/2} \cdot \frac{\tanh mL}{L} \\ \eta &= \frac{\tanh mL}{mL} \quad \dots(3.6.3) \end{aligned}$$

#### 3.6.1 Concept of Corrected Fin Length

For rectangular fin of infinite length;

From Equation (3.6.2)

$$\begin{aligned} \eta &= \frac{\sqrt{h P k A} \cdot \theta_0}{h P L \cdot \theta_0} \\ &= \sqrt{\frac{k \cdot A}{h \cdot P}} \times \frac{1}{L} = \frac{1}{mL} \quad \dots(3.6.4) \end{aligned}$$

Refer Fig. 3.4.1(a). For rectangular fin.

$$P = 2(w + t), \quad A = w \cdot t$$

On substituting in Equation (3.6.4)

$$\eta = \sqrt{\frac{k \cdot A}{h \cdot P}} \cdot \frac{1}{L}$$

$$= \sqrt{\frac{k \cdot w \cdot t}{h \times 2(w+t)}} \times \frac{1}{L} \quad \dots(3.6.5)$$

For a thin fin i.e.  $t \ll w$ . Value of  $t$  can be neglected in  $2(w+t)$  i.e.  $2(w+t) \approx 2w$ . Therefore,

$$\begin{aligned} \eta &= \sqrt{\frac{k \cdot w \cdot t}{2 h w}} \cdot \frac{1}{L} \\ &= \sqrt{\frac{k \cdot t}{2 h}} \cdot \frac{1}{L} \end{aligned}$$

On multiplying and dividing by  $L^{3/2}$  we get,

$$\begin{aligned} \eta &= \sqrt{\frac{k \cdot t}{2 h}} \times \frac{\sqrt{L}}{L^{3/2}} \\ &= \sqrt{\frac{k \cdot t \cdot L}{2 h}} \times \frac{1}{L^{3/2}} \end{aligned}$$

But,  $t \cdot L =$  Profile area of the fin,  $A_m$

$$= \sqrt{\frac{k \cdot A_m}{2 h}} \times \frac{1}{L^{3/2}} \quad \dots(i)$$

On comparing Equation (3.6.4) with Equation (i),

$$mL = \sqrt{\frac{2 h}{k \cdot A_m}} \cdot L^{3/2} \quad \dots(ii)$$

For a real rectangular fin which is long enough, the actual efficiency can be calculated by using Equation (3.6.3) by replacing  $L$  by corrected length  $Le$  such that :

$$Le = L + \frac{t}{2} \quad \dots(3.6.6)$$

This corrected length compensates for the convective heat loss from the tip of real fin. Thus Equation (3.6.3) using Equation (3.6.6) can be rewritten as :

$$\eta = \frac{\tanh \left[ m \times \left( L + \frac{t}{2} \right) \right]}{m \left( L + \frac{t}{2} \right)} \quad \dots(3.6.7)$$

### 3.7 Overall Fin Effectiveness ( $E_0$ )

The overall fin effectiveness,  $E_0$  for a finned surface is defined as :

$$E_0 = \frac{\text{Total heat transfer from finned surface, } Q_{\text{total}}}{\text{Heat transfer from base surface if there were no fins, } Q_{\text{no fins}}} \quad \dots(3.7.1)$$

Let,

- $A_{\text{unfin}}$  = Area of unfinned portion of the wall
- $A_{\text{fin}}$  = Total surface area of all the fins.
- $\eta$  = fin efficiency
- $A_{\text{no fin}}$  = Area of surface when there are no fins.

$$\therefore E_0 = \frac{h (A_{\text{unfin}} + \eta \cdot A_{\text{fin}}) (T_0 - T_{\infty})}{h \cdot A_{\text{no fin}} \cdot (T_0 - T_{\infty})} \quad \dots(3.7.2)$$

where  $A_{\text{fin}}$  is total surface area of all the fins

It is obvious from above equation, the overall effectiveness of the fin will depend on the number of fins per unit length and the efficiency of the fins.

### 3.7.1 Area Weighted Fin Efficiency ( $\eta_{\text{total}}$ )

Practically from a finned surface, the total heat transfer is the sum of heat transfer from the finned surfaces and the heat transfer from the unfinned surface. Thus,

$$\begin{aligned} Q_{\text{total}} &= Q_{\text{fin}} + Q_{\text{unfinned}} \\ &= \eta \times h \cdot A_{\text{fin}} \cdot (T_0 - T_{\infty}) \\ &\quad + (A_{\text{total}} - A_{\text{fin}}) \cdot h (T_0 - T_{\infty}) \quad \dots(3.7.3) \end{aligned}$$

where,  $A_{\text{total}} = A_{\text{unfinned}} + A_{\text{fin}}$

Therefore, we define the total fin efficiency or area weighted fin efficiency,  $\eta_{\text{total}}$  as :

$$\begin{aligned} \eta_{\text{total}} &= \frac{\text{Total heat transfer from finned surface}}{\text{The heat which would be transferred if the total area were maintained at the base temperature, } T_0} \\ &= \frac{\eta \times h \cdot A_{\text{fin}} \cdot (T_0 - T_{\infty}) + (A_{\text{total}} - A_{\text{fin}}) h (T_0 - T_{\infty})}{h \cdot A_{\text{total}} (T_0 - T_{\infty})} \quad \dots(3.7.4) \end{aligned}$$

$$= 1 - \frac{A_{\text{fin}}}{A_{\text{total}}} (1 - \eta) \quad \dots(3.7.5)$$

### 3.8 Error in Temperature Measurement by Thermometer

The temperature of fluid flowing in a duct is often measured by thermometers or thermocouples by placing them into the well, as shown in Fig. 3.8.1.

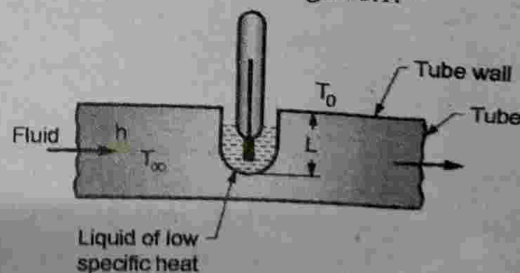


Fig. 3.8.1 : Error in measurement of temperature

The arrangement consists of a hollow tube closed at one end and it is welded into the tube wall of container or pipe such that the cavity is formed. Such a cavity is known as **thermo well**.

The thermometer is placed into the well containing liquid of low specific heat.

Since the thickness of the tube is very small, it can be treated as fin.

### Principle

From temperature distribution of fin, we know that temperature along the length of fin approaches the surrounding temperature and at the end of fin it shall be nearest to the fluid temperature.

However, the temperature at the end of fin will become equal to that of surrounding fluid only in case of fin of infinite length.

Since fin of infinite length is not possible practically, the recorded temperature by thermometer is always less than the fluid temperature. This is known as **error in temperature measurement**.

Considering the well as long fin, the temperature distribution is given by the Equation (3.4.10) as.

$$\frac{\theta}{\theta_0} = \frac{\cosh m(L-x)}{\cosh mL}$$

Temperature at the end of fin i.e. at  $x = L$  becomes :

$$\frac{\theta_L}{\theta_0} = \frac{1}{\cosh mL} \quad \dots(3.8.1)$$

$$\text{where, } m = \sqrt{\frac{h \cdot P}{k \cdot A}}$$

$\theta_L$  = temperature difference at the end =  $T_\infty - T_L$

and  $\theta_0$  = temperature difference at the base =  $T_\infty - T_0$

$$\therefore \frac{\theta_L}{\theta_0} = \frac{T_\infty - T_L}{T_\infty - T_0} = \frac{1}{\cosh mL} \quad \dots(3.8.2)$$

( $T_\infty - T_L$ ) represents the error in temperature measurement. Consider a thermowell like a pin fin of diameter 'd' and thickness 't' Then,

$$P = \pi d \quad \text{and} \quad A = \pi d \cdot t$$

$$\text{i.e. } \frac{P}{A} = \frac{1}{t} \quad (\text{when } d \gg t)$$

$$\therefore m = \sqrt{\frac{h \cdot P}{k \cdot A}} = \sqrt{\frac{h}{k} \times \frac{1}{t}} \quad \dots(3.8.3)$$

From Equations (3.8.2) and (3.8.3) we have,

$$T_\infty - T_L \propto \frac{1}{\cosh mL}$$

As L increases,  $\cosh mL$  decreases.

$$\therefore T_\infty - T_L \propto \frac{1}{mL} \propto \frac{1}{L \times \sqrt{\frac{h}{k} \times \frac{1}{t}}} \propto \sqrt{\frac{k \cdot t}{h}} \cdot \frac{1}{L} \quad \dots(3.8.4)$$

Therefore, in order to minimise error in temperature measurement i.e. ( $T_\infty - T_L$ ) to be minimum we should :

1. Increase the length of thermowell as long as possible. If spacing is less, it can be formed in an inclined position.
2. Thickness of thermowell should be as low as possible.
3. Thermal conductivity of well should be as small as possible.

Table 3.8.1 : List of formulae

### 1. Perimeter and Area of Cross-Section.

Refer Fig. 3.4.1.

(i) Rectangular fins : Length, L

$$P = 2(w + t)$$

If  $t \ll w$  (very thin fin)

$$P = 2w ; \quad A_c = t \cdot w$$

(ii) Pin fin of Length, L

$$P = \pi \cdot d ; \quad A_c = \frac{\pi}{4} d^2$$

### 2. General Equation : $\theta = (T - T_\infty)$

$$\frac{d^2 T}{dx^2} = \frac{h \cdot P}{k \cdot A} (T - T_\infty) = \frac{h \cdot P}{k \cdot A} \cdot \theta = m^2 \cdot \theta$$

$$\theta = C_1 e^{mx} + C_2 e^{-mx}$$

$$\text{where, } m = \sqrt{\frac{h \cdot P}{k \cdot A}}$$

Case I : Infinitely Long fin : Boundary conditions (B.C.) are :

1. At  $x = 0, \quad T = T_0$
2. At  $x = L \rightarrow \infty, \quad T = T_\infty$

Temperature distribution :

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{-mx} \quad \dots(3.4.7)$$

$$Q = -k \cdot A \left( \frac{d\theta}{dx} \right)_{x=0} = \sqrt{k \cdot A \cdot h \cdot P} (T_0 - T_\infty) \quad \dots(3.4.8)$$

Case II : Adequately Long fin with insulated end :

B.C. are :

1. At  $x = 0, \quad T = T_0$  i.e.  $\theta = \theta_0$
2. At  $x = L, \quad Q = 0$  i.e.  $\frac{d\theta}{dx} = 0$

Temperature distribution :

$$\frac{\theta}{\theta_0} = \frac{T - T_\infty}{T_0 - T_\infty} = \frac{\cosh m(L-x)}{\cosh mL} \quad \dots(3.4.10)$$

$$\text{Heat flow rate, } Q = \sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \tanh mL \quad \dots(3.4.12)$$



where,  $m = \sqrt{\frac{h \cdot P}{k \cdot A}}$

Case III : Short fin

B.C. are :

1. At  $x = 0$ ,  $T = T_0$  and  $\theta = \theta_0$
2. At  $x = L$ ,  $Q = Q_c$

Temperature distribution :

$$\frac{\theta}{\theta_0} = \frac{\cosh m(L-x) + \left(\frac{h}{mk}\right) \sinh m(L-x)}{\cosh mL + \left(\frac{h}{mk}\right) \sinh mL} \quad \dots(3.4.15)$$

Heat flow rate,  $Q$  :

$$Q = \sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \left[ \frac{\tanh(mL) + \left(\frac{h}{mk}\right)}{1 + \left(\frac{h}{mk}\right) \tanh(mL)} \right] \quad \dots(3.4.17)$$

**Important :** Find  $mL$ . If  $\tanh mL \geq 0.99$  it is considered infinitely long fin.

### 3. Effectiveness of fin, $E$

$$E = \frac{\text{Actual heat transfer from finned surface, } Q}{\text{Rate of heat transfer without fins}}$$

For fin with insulated tip,

$$E = \tanh(mL) \times \sqrt{\frac{P \cdot k}{h \cdot A}} = \sqrt{\frac{k \cdot P}{h \cdot A}} \quad [\text{if } mL \geq 3]$$

### 4. Fin Efficiency, $\eta$

$$\frac{Q}{Q_{\max}} = \frac{Q}{h \cdot A_f \cdot \theta_0} = \frac{Q}{h \cdot P \cdot L \cdot \theta_0}$$

(where,  $A_f = P \cdot L$ )

(a) For long fin with insulated tip

$$\eta = \frac{\tanh mL}{mL} \quad \dots(3.6.3)$$

(b) For infinitely long fin

$$\eta = \frac{1}{mL} \quad \dots(3.6.4)$$

### 5. Overall fin Effectiveness

$$E_0 = \frac{h(A_{\text{unfin}} + \eta \cdot A_{\text{fin}})(T_0 - T_{\infty})}{h \cdot A_{\text{no fin}}(T_0 - T_{\infty})} \quad \dots(3.7.2)$$

### 6. Area weighted fin Efficiency, $\eta_{\text{total}}$

$$\eta_{\text{total}} = 1 - \frac{A_{\text{fin}}}{A_{\text{total}}}(1 - \eta) \quad \dots(3.7.5)$$

### 7. Error in temperature measurement

$$\frac{\theta_L}{\theta_0} = \frac{T_{\infty} - T_L}{T_{\infty} - T_0} = \frac{1}{\cosh mL} \quad \dots(3.8.2)$$

For thermowell :

$$P = \pi \cdot d; \quad A = \pi \cdot d \cdot l$$

$$m = \sqrt{\frac{h \cdot P}{k \cdot A}} = \sqrt{\frac{h}{k \cdot l}}$$

$$\therefore (T_{\infty} - T_L) \propto \frac{1}{mL} \propto \sqrt{\frac{k \cdot l}{h}} \cdot \frac{1}{L}$$

**Ex. 3.8.1 :** A cylindrical metal rod of 5 cm diameter and 20 cm long with thermal conductivity 225 W/mK protrudes in atmosphere at 30°C. It projects from furnace wall at 300°C. A convective heat transfer coefficient of air is 10 W/m<sup>2</sup>K. Determine temperature at the free end of the rod assuming it as a fin insulated at end.

**Soln. :**

**Given :**  $d = 5 \text{ cm} = 0.05 \text{ m}$

$$A = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times (0.05)^2 = 1.963 \times 10^{-3}$$

$$l = 20 \text{ cm} = 0.2 \text{ m}$$

$$k = 225 \text{ W/mK,}$$

$$h = 10 \text{ W/m}^2 \text{ K}$$

$$T_{\infty} = 30^\circ \text{C}; \quad T_b = 300^\circ \text{C};$$

$$\theta_0 = T_b - T_{\infty} = (300 - 30) = 270^\circ \text{C}$$

**Fin tip is insulated :**

$$m = \sqrt{\frac{Ph}{kA}} = \sqrt{\frac{\pi \cdot d \times h}{k \times \frac{\pi}{4} d^2}} = \sqrt{\frac{4h}{k \cdot d}}$$

$$= \sqrt{\frac{4 \times 10}{225 \times 0.05}} = 1.88$$

$$\text{Heat transfer rate, } Q = m A k \theta_0 \tanh mL$$

$$Q = 1.88 \times 1.963 \times 10^{-3} \times 225 \times (270) \times \tanh(1.88 \times 0.2)$$

$$= 80.55 \text{ W}$$

**Tip temperature,  $T_f$  or  $T_L$  at  $x = L$**

$$\frac{\theta_L}{\theta_0} = \frac{\cosh m(L-x)}{\cosh mL}$$

$$\frac{T_L - T_{\infty}}{T_b - T_{\infty}} = \frac{1}{\cosh mL}$$

$$\frac{T_L - 30}{100 - 30} = \frac{1}{\cosh \times (1.88 \times 0.2)} = 0.933$$

$$T_L = 281.91^\circ \text{C}$$

...Ans.

**Ex. 3.8.2 :** A cylindrical fin is 3 mm in diameter and 3 cm long. Calculate the value of the temperature at fin tip if the fin is made of

(i) Copper ( $k = 350 \text{ W/mK}$ ) and

(ii) Teflon ( $K = 0.35 \text{ W/mK}$ ).

Assume that heat loss from fin tip is negligible.

Take  $h = 10 \text{ W/m}^2 \text{ K}$ ,  $T_f = 20^\circ \text{C}$ ,  $T_{\text{base}} = 120^\circ \text{C}$ .

**Soln. :**

$$d = 3\text{ mm} = 0.003\text{ m,}$$

$$l = 3\text{ cm} = 0.03\text{ m}$$

**(i) Copper fin**

Fin tip temperature shall be calculated as,

$$\frac{\theta}{\theta_b} = \frac{\cosh m(l-x)}{\cosh ml}$$

for negligible heat loss from fin tip

For tip temperature, putting  $x = l$ 

$$\frac{\theta_l}{\theta_b} = \frac{1}{\cosh ml}$$

$$m = \sqrt{\frac{Ph}{kA}} = \sqrt{\frac{\pi d \times h}{k \times \frac{\pi}{4} d^2}}$$

$$= \sqrt{\frac{4h}{kd}} = \sqrt{\frac{4 \times 10}{350 \times 0.003}}$$

$$= 6.172$$

$$\theta_l = \frac{120 - 20}{\cosh(6.172 \times 0.03)} = \frac{100}{1.0172} = 98.31$$

$$T_l - T_f = 98.31$$

$$T_l = 118.3^\circ\text{C}$$

...Ans.

**(ii) Teflon fin**

$$m = \sqrt{\frac{4h}{kd}} = \sqrt{\frac{4 \times 10}{0.35 \times 0.003}}$$

$$= 195.2$$

$$T_l - T_f = \frac{120 - 20}{\cosh(195.2 \times 0.03)} = 0.57$$

$$T_l - T_f = 0.57$$

$$T_l = 20.57^\circ\text{C}$$

...Ans.

**Ex 3.8.3 :** An aluminium rod 2.0 cm diameter and 10 cm long protrudes from the wall maintained at  $300^\circ\text{C}$ . The rod is exposed to surroundings at  $15^\circ\text{C}$ . Heat transfer coefficient between rod surface and environment is  $20\text{ W/m}^2\text{K}$ . The thermal conductivity of the material is  $200\text{ W/mK}$ . Find :

- Total heat dissipated by rod
- Temperature of rod at 4 cm from the wall.
- Temperature at the end of rod.
- Fin efficiency.

Assume that the rod end is insulated.

$$\text{Soln. : } d = 2\text{ cm} = 0.02\text{ m,}$$

$$L = 10\text{ cm} = 0.1\text{ m,}$$

$$T_0 = 300^\circ\text{C}$$

$$T_\infty = 15^\circ\text{C}$$

$$h = 20\text{ W/m}^2\text{K}$$

$$k = 200\text{ W/mK.}$$

$$\therefore \theta_0 = T_0 - T_\infty = 300 - 15 = 285^\circ\text{C}$$

$$P = \pi d = \pi \times 0.02 = 0.06283\text{ m}$$

$$A = \frac{\pi}{4} d^2 = \frac{\pi}{4} \times (0.02)^2 = 0.0003142\text{ m}^2$$

$$m = \sqrt{\frac{h \cdot P}{k \cdot A}} = \sqrt{\frac{20 \times 0.06283}{200 \times 0.0003142}} = 4.472$$

This problem is based on adequately long fin with insulated end i.e. (Case II)

**(i) Total heat dissipated by rod, Q**

$$Q = \sqrt{h P k A} \cdot \theta_0 \cdot \tanh mL$$

... (From Equation 2.8.12)

$$= \sqrt{20 \times 0.06283 \times 200 \times 0.0003142} \times 285 \times \tanh(4.472 \times 0.1)$$

$$Q = 33.6\text{ W}$$

...Ans.

**(ii) Temperature of rod at 4cm from the wall,**i.e. T at  $x = 0.04\text{ m}$ 

$$\frac{T - T_\infty}{T_0 - T_\infty} = \frac{\cosh m(L-x)}{\cosh mL}$$

$$\frac{T - 15}{300 - 15} = \frac{\cosh [4.472(0.1 - 0.04)]}{\cosh [4.472 \times 0.1]}$$

$$\frac{T - 15}{285} = \frac{1.0362}{1.1017}$$

$$T = 283.06^\circ\text{C}$$

...Ans.

**(iii) Temperature at the end of rod**i.e. T at  $x = L = 0.1\text{ m}$ 

$$\frac{T - T_\infty}{T_0 - T_\infty} = \frac{\cosh m(L-x)}{\cosh mL}$$

$$\frac{T - 15}{300 - 15} = \frac{\cosh [4.472 \times (0.1 - 0.1)]}{\cosh [4.472 \times 0.1]}$$

$$T = 273.7^\circ\text{C}$$

...Ans.

**(iv) Fin efficiency,  $\eta$  :**

$$\eta = \frac{\tanh mL}{mL}$$

$$= \frac{\tanh(4.472 \times 0.1)}{4.472 \times 0.1}$$

$$= 0.9383\text{ or }93.83\%$$

...Ans.

**Ex. 3.8.4 :** An electric motor is to be connected by a horizontal steel shaft ( $k = 42.56 \text{ W/mK}$ ), 25 mm in diameter to an impeller of a pump, circulating liquid metal at a temperature of  $540^\circ\text{C}$ . If the temperature of electric motor is limited to a maximum value of  $52^\circ\text{C}$  with the ambient air at  $27^\circ\text{C}$  and heat transfer coefficient of  $40.7 \text{ W/m}^2\text{K}$ , what length of shaft should be specified between motor and pump? Assume insulated tip condition, for fin analysis.

**Soln. :**

Refer Fig. P. 3.8.4

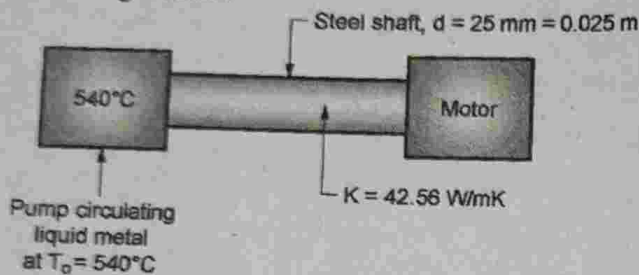


Fig. P. 3.8.4

**Given :**  $d = 25 \text{ mm} = 0.025 \text{ m}$ ,  $T_{\infty} = 27^\circ\text{C}$ ,  
 $k = 42.56 \text{ W/mK}$   
 $T_0 = 540^\circ\text{C}$ ,  $T_L = 52^\circ\text{C}$ ,  
 $h = 40.7 \text{ W/m}^2\text{K}$

$\therefore$  One end of shaft is connected to motor, therefore assuming no heat loss from fin tip.

The temperature distribution, treating the shaft as a fin insulated at the tip is given by

$$\frac{\theta}{\theta_0} = \frac{T_L - T_{\infty}}{T_0 - T_{\infty}} = \frac{\cosh [m(L-x)]}{\cosh (mL)}$$

At  $x = L$ ,  $T = T_L$  the above equation reduces to

$$\frac{T_L - T_{\infty}}{T_0 - T_{\infty}} = \frac{1}{\cosh (mL)}$$

$$\frac{52 - 27}{540 - 27} = \frac{1}{\cosh (mL)}$$

$$\therefore \cosh (mL) = 20.52$$

$$\therefore (mL) = 3.714$$

$$\text{But } m = \sqrt{\frac{hP}{kA}} = \sqrt{\frac{h \times \pi d}{k \times \frac{\pi}{4} d^2}}$$

$$= \sqrt{\frac{4h}{kd}} = \sqrt{\frac{4 \times 40.7}{42.56 \times 0.025}}$$

$$= 12.37$$

$$\text{But, } mL = 3.714$$

$$12.37 L = 3.714$$

$$\therefore L = 0.300 \text{ m}$$

$$\therefore L = 300 \text{ mm}$$

...Ans.

**Ex. 3.8.5 :** A copper rod 0.5 cm diameter and 50 cm long protrudes from a wall maintained at a temperature of  $500^\circ\text{C}$ . The surrounding temperature is  $30^\circ\text{C}$ . Convective heat transfer coefficient is  $40 \text{ W/m}^2\text{K}$  and thermal conductivity of material is  $300 \text{ W/m K}$ . Determine :

- Total heat transfer rate from rod
- Temperature of the rod at 20 cm from wall.

**Soln. :**

$$d = 0.5 \text{ cm} = 0.005 \text{ m}, \quad L = 50 \text{ cm} = 0.5 \text{ m}$$

$$T_0 = 500^\circ\text{C},$$

$$T_{\infty} = 30^\circ\text{C}$$

$$h = 40 \text{ W/m}^2\text{K}$$

$$k = 300 \text{ W/m K.}$$

$$\therefore P = \pi \cdot d = \pi \times 0.005 = 15.7 \times 10^{-3} \text{ m}$$

$$A = \frac{\pi}{4} \cdot (d)^2 = \frac{\pi}{4} \times (0.005)^2 = 1.9634 \times 10^{-5} \text{ m}^2$$

$$m = \sqrt{\frac{h \cdot P}{k \cdot A}} = \sqrt{\frac{40 \times 15.7 \times 10^{-3}}{300 \times 1.9634 \times 10^{-5}}} = 10.325$$

$$mL = 10.325 \times 0.5 = 5.163$$

Therefore, fin can be assumed to be infinity long fin since  $mL > 3$ .

- Total heat transfer from fin,  $Q$  :

$$Q = \sqrt{k \cdot A \cdot h \cdot P} (T_0 - T_{\infty})$$

...(from Equation 3.4.8)

$$= \sqrt{300 \times (1.9634 \times 10^{-5}) \times 40 \times (15.7 \times 10^{-3})} \times (500 - 30)$$

$$= 28.585 \text{ W}$$

...Ans.

- Temperature of rod at  $x = 20 \text{ cm} = 0.2 \text{ m}$  from wall,  $T$

$$\frac{T - T_{\infty}}{T_0 - T_{\infty}} = e^{-mx}$$

$$\frac{(T - 30)}{(500 - 30)} = e^{-10.325 \times 0.2}$$

$$T = 89.6^\circ\text{C}$$

...Ans.

**Ex. 3.8.6 :** Following are the specifications of an insulated end, circular cross sectioned fin.

Length = 25 cm ; Diameter = 2.5 cm ; Base surface temperature =  $100^\circ\text{C}$ .

Heat transfer coefficient between fin surface and surrounding air =  $20 \text{ W/m}^2\text{K}$

Temperature of surrounding air =  $30^\circ\text{C}$ . Calculate the tip temperature of the fin if it is made up of,

- copper with  $k = 380 \text{ W/m K}$
- brass with  $k = 110 \text{ W/m K}$

**Soln. :**

**Given :**  $L = 25 \text{ cm} = 0.25 \text{ m}$ ,  $d = 2.5 \text{ cm} = 0.025 \text{ m}$ ,  
 $T_0 = 100^\circ\text{C}$ ,  $h = 20 \text{ W/m}^2 \text{ K}$ ,  $T_\infty = 30^\circ\text{C}$

Circular fin;  $P = \pi \cdot d$  and  $A = \frac{\pi}{4} \cdot d^2$

$$m = \sqrt{\frac{h \cdot P}{k \cdot A}} = \sqrt{\frac{h \cdot \pi \cdot d}{k \cdot (\pi/4) d^2}} = \sqrt{\frac{4h}{k \cdot d}} \quad \dots(i)$$

(i) Temperature at the end of rod,

T i.e. at  $x = L = 0.2$  if  $k = 380 \text{ W/m K}$  :

$$\frac{\theta_L}{\theta_0} = \frac{T_x - T_\infty}{T_0 - T_\infty} = \frac{\cosh m(L-x)}{\cosh mL}$$

$$\text{At } x = L, \frac{T_L - T_\infty}{T_0 - T_\infty} = \frac{1}{\cosh mL} \quad \dots(ii)$$

$$m = \sqrt{\frac{4 \times 20}{380 \times 0.025}} = 2.90$$

$$\therefore \frac{T_L - 30}{100 - 30} = \frac{1}{\cosh (2.9 \times 0.25)} = \frac{1}{1.27453} = 0.7846$$

$$T_L = 84.92^\circ\text{C} \quad \dots\text{Ans.}$$

(ii) Temperature at the end of rod,

$T_L$  at  $x = L$  if  $k = 110 \text{ W/m}$

From Equation (i) :

$$m = \sqrt{\frac{4 \cdot h}{k \cdot d}} = \sqrt{\frac{4 \times 20}{110 \times 0.025}} = 5.394$$

From Equation (ii) :

$$\frac{T_L - 30}{100 - 30} = \frac{1}{\cosh (5.394 \times 0.25)} = 0.4865$$

$$T_L = 64.06^\circ\text{C} \quad \dots\text{Ans.}$$

**Ex. 3.8.7 :** Two rods of identical size and shape are both supported on a heat source at  $100^\circ\text{C}$  and are surrounded by air at  $25^\circ\text{C}$ . One rod is known to have a thermal conductivity of  $43 \text{ W/m-K}$  and its midpoint temperature is measured to be  $49^\circ\text{C}$ . If the midpoint temperature of the other rod is measured to be  $75^\circ\text{C}$ , what is its thermal conductivity ?

**Soln. :**

Refer Fig. P. 3.8.7.

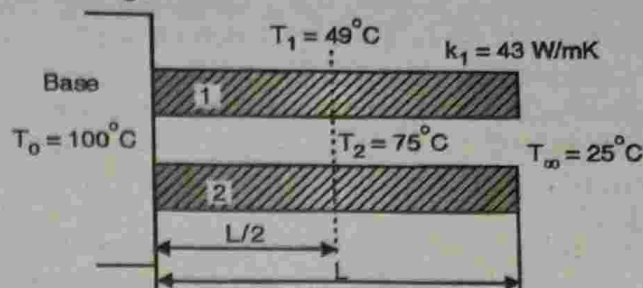


Fig. P. 3.8.7 : Two identical rod from a base

The temperature distribution in the fin is given by the Equation (3.4.7) as :

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{-mx}$$

$$T = T_\infty + (T_0 - T_\infty) \cdot e^{-mx}$$

For rod 1 :

$$49 = 25 + (100 - 25) e^{(-m_1 \cdot \frac{L}{2})} \quad \dots(i)$$

For rod 2 :

$$75 = 25 + (100 - 25) e^{(-m_2 \cdot \frac{L}{2})} \quad \dots(ii)$$

From Equation (i) :

$$e^{(-m_1 \cdot \frac{L}{2})} = 0.32$$

$$m_1 L = 2.2789 \quad \dots(iii)$$

From Equation (ii)

$$e^{(-m_2 \cdot \frac{L}{2})} = 0.67$$

$$m_2 L = 0.8109 \quad \dots(iv)$$

On dividing Equation (iii) to (iv),

$$\frac{m_1}{m_2} = \frac{2.2789}{0.8109} = 2.8102$$

Since rods are of identical size, therefore

$$P_1 = P_2, A_1 = A_2 \text{ and } h_1 = h_2$$

$$\frac{m_1}{m_2} = \frac{\sqrt{\frac{hP}{k_1 A}}}{\sqrt{\frac{hP}{k_2 A}}} = 2.8102$$

$$\therefore \frac{k_2}{k_1} = 7.8974, \text{ but } k_1 = 43 \text{ W/mK}$$

$$k_2 = 339.56 \text{ W/mK} \quad \dots\text{Ans.}$$

**Ex. 3.8.8 :** The end of a very long cylindrical stainless steel rod is attached to a heated wall and its surface is in contact with a cold fluid. Determine by what percentage the heat removal rate would change.

- (i) If the rod diameter were doubled.  
(ii) If the rod is made up of aluminium.

Use  $k$  for  $Al = 204.7 \text{ W/mK}$ ;  $k$  for S.S. =  $16.17 \text{ W/mK}$ .

**Soln. :**

For fin of infinite length, heat transferred  $Q$ ,

$$Q = \sqrt{k \cdot A \cdot h \cdot P} (T_0 - T_\infty) \quad \dots(\text{from Equation 3.4.8})$$

- (i) Percentage change in heat removal rate if rod diameter is doubled, i.e.,

$$d_2 = 2d_1$$

Cross sectional area,

$$A_1 = \frac{\pi}{4} d_1^2; A_2 = \frac{\pi}{4} (2d_1)^2 = 4 \left( \frac{\pi}{4} d_1^2 \right) = 4A_1$$

$$\text{Perimeter, } P_1 = \pi d_1; P_2 = \pi d_2 = \pi \times 2d_1 = 2P_1$$

Using Equation (2.8.8), we can write,

$$\frac{Q_1}{Q_2} = \frac{\sqrt{h \cdot P_1 \cdot k \cdot A_1}}{\sqrt{h \cdot P_2 \cdot k \cdot A_2}} \times \frac{(T_0 - T_\infty)}{(T_0 - T_\infty)}$$

$$\frac{Q_1}{Q_2} = \sqrt{\frac{P_1 \cdot A_1}{P_2 \cdot A_2}} = \sqrt{\frac{P_1}{2P_1} \times \frac{A_1}{4A_1}} = \sqrt{\frac{1}{8}}$$

$$Q_2 = 2\sqrt{2} Q_1$$

$$\therefore \% \text{ change in heat removal rate} = \frac{Q_2 - Q_1}{Q_1} \times 100 \%$$

$$= \frac{(2\sqrt{2} \cdot Q_1 - Q_1)}{Q_1} \times 100$$

$$= 182.84\%$$

...Ans.

- (ii) Percentage change in heat removal rate if the stainless steel rod is replaced by aluminium rod.

For stainless steel fin,  $k = k_1 = 16.17 \text{ W/mK}$

$$\therefore Q_1 = \sqrt{h P k_1 A} (T_0 - T_\infty)$$

$$= \sqrt{16.17} \sqrt{h P A} (T_0 - T_\infty)$$

$$Q_1 = 4.0212 \sqrt{h P A} (T_0 - T_\infty)$$

For Aluminum fin,  $k_{Al} = k_2 = 204.7 \text{ W/mK}$

$$Q_2 = \sqrt{h P k_2 A} (T_0 - T_\infty)$$

$$= \sqrt{204.7} \sqrt{h P A} (T_0 - T_\infty)$$

$$Q_2 = 14.307 \sqrt{h P A} (T_0 - T_\infty)$$

Percentage change in heat removal rate

$$= \left( \frac{Q_2 - Q_1}{Q_1} \right) \times 100 \%$$

$$= \frac{(14.307 - 4.0212)}{4.0212} \times 100$$

$$= 255.798\%$$

...Ans.

**Ex. 3.8.9** A steel rod ( $k = 30 \text{ W/m}^\circ\text{C}$ ) 1 cm in diameter and 5 cm long protrudes from a wall which is maintained at  $100^\circ\text{C}$ . The rod is exposed to an environment with temperature  $30^\circ\text{C}$  and  $h = 50 \text{ W/m}^2\text{K}$ . Calculate fin efficiency, temperature at the tip of fin and rate of heat dissipation from the base of the fin. Assume insulated tip condition, for fin analysis.

**Soln. :**

$$\text{Given: } k = 30 \text{ W/m}^\circ\text{C}; \quad d = 1 \text{ cm} = 0.01 \text{ m}$$

$$l = 5 \text{ cm} = 0.05 \text{ m}$$

$$T_b = 100^\circ\text{C}$$

$$T_\infty = 30^\circ\text{C}$$

$$h = 50 \text{ W/m}^2\text{K}$$

**Fin tip is insulated**

$$\text{Heat transfer rate, } Q = m A k Q_b \tanh ml$$

$$m = \sqrt{\frac{Ph}{kA}} = \sqrt{\frac{\pi \cdot d \times h}{k \times \frac{\pi}{4} d^2}}$$

$$= \sqrt{\frac{4h}{k \cdot d}} = \sqrt{\frac{4 \times 50}{30 \times 0.01}} = 25.82$$

$$Q = 25.82 \times \frac{\pi}{4} \times 0.01^2 \times 30 \times (100 - 30) \times \tanh(25.82 \times 0.05) = 3.66 \text{ W}$$

- (i) Tip temperature,  $T_e$

$$\frac{Q}{Q_b} = \frac{\cosh m(l-x)}{\cosh ml}$$

$$\frac{T_e - T_\infty}{T_b - T_\infty} = \frac{1}{\cosh ml}$$

$$\frac{T_e - 30}{100 - 30} = \frac{1}{\cosh \times (25.82 \times 0.05)} = 0.511$$

$$T_e = 65.8^\circ\text{C}$$

...Ans.

- (ii) Fin efficiency

$$\eta = \frac{\tanh ml}{ml} = \frac{\tanh(25.82 \times 0.05)}{25.82 \times 0.05}$$

$$= 0.666 \text{ or } 66.6 \%$$

...Ans.

(iii)  $Q_{\text{base}}$

$$Q = hA Q_b = 50 \times \frac{\pi}{4} \times 0.01^2 \times (100 - 30) \\ = 0.2748 \text{ W} \quad \dots \text{Ans.}$$

**Ex. 3.8.10 :** Pin fins are provided on the surfaces to increase the heat transfer rates. Two arrangements are possible as given below :

(i) 5 fins of 10 cm length.

(ii) 10 fins of 5 cm length

Assume :  $k = 300 \text{ W/mK}$ ,  $h = 15 \text{ W/m}^2 \text{ K}$ , cross-sectional area of each fin,  $A_c = 3 \text{ cm}^2$ , Perimeter of each fin,  $P = 6 \text{ cm}$ . Surface temperature,  $T_0 = 250^\circ\text{C}$ , Surrounding air temperature,  $T_\infty = 30^\circ\text{C}$ . Ends of the fins are assumed to be insulated. Find which of the above two arrangements will be more effective for transfer of heat ?

**Soln. :**

$$A = 3 \text{ cm}^2 = 3 \times 10^{-4} \text{ m}^2 \\ \text{and } P = 6 \text{ cm} = 6 \times 10^{-2} \text{ m}$$

$$m = \sqrt{\frac{h \cdot P}{k \cdot A}} \\ = \sqrt{\frac{15 \times (6 \times 10^{-2})}{300 \times (3 \times 10^{-4})}} = 3.1623 \\ \theta_0 = T_0 - T_\infty \\ = 250 - 30 = 220^\circ\text{C}$$

(i) Heat transfer,  $Q_1$  from 5 fins of 10 cm length i.e.  $L_1 = 0.1 \text{ m}$

$$Q_1 = \text{No. of fins} \times \sqrt{h \cdot P \cdot k \cdot A} \times \theta_0 \\ \times \tanh(mL_1) \quad \dots (1)$$

$$Q_1 = 5 \times \sqrt{15 \times (6 \times 10^{-2}) \times 300 \times (3 \times 10^{-4})} \\ \times 220 \times \tanh(3.1623 \times 0.1)$$

$$Q_1 = 95.83 \text{ W} \quad \dots \text{Ans.}$$

(ii) Heat transfer,  $Q_2$  from 10 fins of 5 cm length i.e.  $L_2 = 0.05 \text{ m}$

$$Q_2 = \text{No. of fins} \times \sqrt{h \cdot P \cdot k \cdot A} \times \theta_0 \\ \times \tanh(mL_2)$$

$$Q_2 = 10 \times \sqrt{15 \times (6 \times 10^{-2}) \times 300 \times (3 \times 10^{-4})} \\ \times 220 \times \tanh(3.1623 \times 0.05)$$

$$Q_2 = 98.18 \text{ W} \quad \dots \text{Ans.}$$

$\therefore Q_2 > Q_1$ , the second arrangement of 10 fins of 5 cm length is more effective for heat transfer.  $\dots \text{Ans.}$

**Ex. 3.8.11 :** Consider a stainless spoon ( $k = 15.1 \text{ W/m}^\circ\text{C}$ ), partially immersed in the boiling water at  $95^\circ\text{C}$  in a kitchen at  $25^\circ\text{C}$ . The handle of the spoon has a cross section  $0.2 \text{ cm} \times 1 \text{ cm}$  and it extends 18 cm in the air from the free surface of the water. If the heat transfer coefficient on the exposed surfaces of the spoon is  $15 \text{ W/m}^2\text{K}$ , Calculate the temperature difference across the exposed surface of the spoon handle. For the analysis, assume fin insulated tip condition.

**Soln. :**

$$k = 15.1 \text{ W/m}^\circ\text{C}$$

$$T_\infty = 25^\circ\text{C}$$

$$T_0 = 95^\circ\text{C}$$

$$\text{Area, } A = 0.2 \text{ cm} \times 1 \text{ cm} = 0.2 \text{ cm}^2 = 0.2 \times 10^{-4} \text{ m}^2$$

$$\text{Perimeter } P = 2(w + t) = 2(0.2 + 1) \\ = 2.4 \text{ cm} = 0.024 \text{ m}$$

$$L = 18 \text{ cm} = 0.18 \text{ m}$$

$$h = 15 \text{ W/m}^2 \text{ K}$$

Temperature distribution for fin with insulated tip is given by,

$$\frac{\theta_L}{\theta_0} = \frac{T_L - T_\infty}{T_0 - T_\infty} = \frac{\cosh m(L - x)}{\cosh mL}$$

$$m = \sqrt{\frac{hP}{kA}} = \sqrt{\frac{15 \times 0.024}{15.1 \times 0.2 \times 10^{-4}}} \\ = 34.52 \text{ at } x = L$$

$$\frac{T_L - T_\infty}{T_0 - T_\infty} = \frac{1}{\cosh mL}$$

$$\frac{T_L - 25}{95 - 25} = \frac{1}{\cosh(34.52 \times 0.18)}$$

$$\therefore \frac{T_L - 25}{95 - 25} = \frac{1}{249.75}$$

$$T_L - 25 = 0.28$$

$$\therefore T_L = 25.28]$$

$$\therefore T_0 - T_L = 95 - 25.28$$

$$= 69.72^\circ\text{C} \quad \dots \text{Ans.}$$

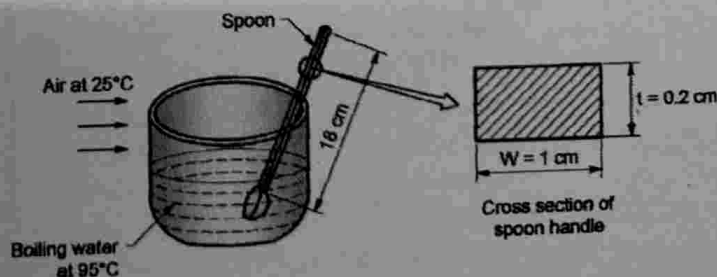


Fig. P. 3.8.11

**Ex. 3.8.12 :** A vertical cylinder ( $k = 55.7 \text{ W/mK}$ ) is fitted with 20 longitudinal fins. (50 mm long and 3 mm thick). Height of the cylinder is 120 cm and its outer diameter is 6 cm. Temperature at the outer surface of the cylinder is  $80^\circ\text{C}$ . Surrounding temperature and heat transfer coefficient are  $18^\circ\text{C}$  and  $9 \text{ W/m}^2\text{K}$  respectively. Assuming insulated tip fins, calculate heat transfer rate from the cylinder along with the fins.

**Soln. :**

$$L = 50 \text{ mm} = 0.05 \text{ m}, \quad t = 3 \text{ mm} = 0.003 \text{ m},$$

$$\text{No. of fins} = 20, \quad k = 55.7 \text{ W/mK},$$

$$\text{Height of cylinder, } l = 120 \text{ cm} = 1.2 \text{ m},$$

$$D = 6 \text{ cm} = 0.06 \text{ m} \quad T_0 = 80^\circ\text{C},$$

$$T_\infty = 18^\circ\text{C}, \quad h = 9 \text{ W/m}^2\text{K}.$$

Refer Fig. P. 3.8.12

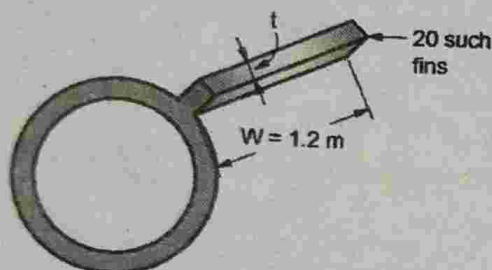


Fig. P. 3.8.12

$$\begin{aligned} \text{Total root area of 20 fins} &= \text{No. of fins} \times l \times t \\ &= 20 \times 1.2 \times 0.003 = 0.072 \text{ m}^2 \end{aligned}$$

Bare area of cylinder

$$\begin{aligned} &= \text{Surface area of cylinder} - \text{Root area of fins} \\ &= \pi d l - \text{Root area of fins} \\ &= \pi \times 0.06 \times 1.2 - 0.072 \\ &= 0.1542 \text{ m}^2 \end{aligned}$$

Heat transfer from bare area,

$$\begin{aligned} Q_1 &= h \cdot A \cdot (T_0 - T_\infty) \\ &= 0.1542 \times 9 \times (80 - 18) = 86 \text{ Watts} \end{aligned}$$

$$\begin{aligned} \text{for fins, } m &= \sqrt{\frac{hP}{kA}} = \sqrt{\frac{9 \times 2(1.2 + 0.003)}{55.7 \times 1.2 \times 0.003}} \\ &= 10.4 \end{aligned}$$

$$mL = 10.4 \times 0.05 = 0.52$$

$$\tanh mL = 0.4777$$

Heat transfer from fins,

$$\begin{aligned} Q_2 &= \sqrt{hPkA} \theta_0 \tanh mL \times \text{No. of fins} \\ &\quad (\text{Where, } \theta_0 = T_0 - T_\infty) \\ &= \sqrt{9 \times 2(1.2 + 0.003) \times 55.7 \times 1.2 \times 0.003} \\ &\quad \times (80 - 18) \times 0.4777 \times 20 \end{aligned}$$

$$= 1234.3 \text{ Watts.}$$

$$\text{Total heat transfer} = Q_2 + Q_1 = 1234.3 + 86$$

$$= 1320.3 \text{ Watts}$$

...Ans.

**Ex. 3.8.13 :** Both ends of a 6 mm diameter 'U' shaped copper rod are rigidly fixed to vertical wall. The wall is maintained at  $100^\circ\text{C}$ . The developed length of the rod is 50 cm and conductivity is  $300 \text{ W/mK}$ . It is exposed to air at  $30^\circ\text{C}$ . The combined convective and radiative heat transfer coefficient is  $30 \text{ W/m}^2\text{K}$ . Calculate (a) Temperature at the centre of the rod. (b) Heat transferred by the rod.

**Soln. :** Refer Fig. P. 3.8.13

$$\text{Given : } d = 6 \text{ mm} = 0.006 \text{ m},$$

$$k = 300 \text{ W/mK},$$

$$T_0 = 100^\circ\text{C},$$

$$T_\infty = 30^\circ\text{C}$$

$$h = 30 \text{ W/m}^2\text{K}$$

$$L = 50 \text{ cm} = 0.5 \text{ m}$$

The heat transfer across the central section must be zero. Hence will consider the rod as fin of length 25 cm each which are insulated at their ends.

(i) Temperature at the centre of rod

$$\begin{aligned} m &= \sqrt{\frac{hP}{kA}} \\ &= \sqrt{\frac{30 \times \pi (0.006)}{300 \times \frac{\pi}{4} (0.006)^2}} \left[ (P = \pi d); A = \frac{\pi}{4} d^2 \right] \\ &= 8.165 \end{aligned}$$

$$L = 50 \text{ cm}, L_1 = \frac{50}{2} = 25 \text{ cm} = 0.25 \text{ m}$$

For fin with insulated tip we have

$$\frac{\theta_L}{\theta_0} = \frac{1}{\cosh(mL_1)}$$

$$\therefore \frac{T_{L1} - T_\infty}{T_0 - T_\infty} = \frac{1}{\cosh(mL_1)}$$

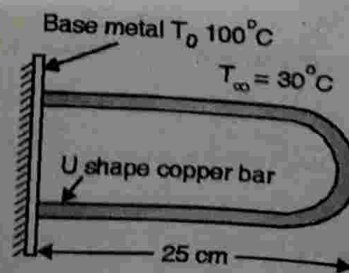


Fig. P. 3.8.13 : U shape copper bar

where  $T_{L1}$  is the temp at end of rod (can be of U shape)

$$\frac{T_{L1} - 30}{100 - 30} = \frac{1}{\cosh(8.16 \times 0.25)}$$

$$T_{L1} = 47.87^\circ\text{C}$$

...Ans.

**(ii) Heat transfer from rod**

$$\begin{aligned}\theta_0 &= T_0 - T_\infty \\ &= 100 - 30 = 70^\circ\text{C}\end{aligned}$$

$$\begin{aligned}Q &= 2 \cdot \sqrt{h P k A} \theta_0 \tanh(mL) \\ &= 2 \cdot \sqrt{30 \times \pi \times (0.006) \times 300 \times \left[\frac{\pi}{4} \times (0.006)^2\right]} \\ &\quad \times 70 \times \tanh(8.165 \times 0.25) \\ &= 9.37 \text{ Watts}\end{aligned}$$

...Ans.

**Ex. 3.8.14 :** Three 10 mm diameter rods A, B and C protrude from a furnace at  $100^\circ\text{C}$ . The length of all of them = 25 cm. They protrude out in ambient air at  $20^\circ\text{C}$ . The temperatures at the other ends are  $26.7^\circ\text{C}$  for 'A',  $32^\circ\text{C}$  for 'B', and  $36.93^\circ\text{C}$  for 'C'. Considering rods to be very long fins, evaluate their thermal conductivities if heat transfer coefficient =  $23 \text{ W/m}^2 \text{ K}$ .

**Soln. :** Refer P. 3.8.14

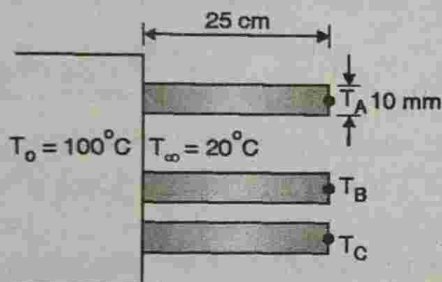


Fig. P. 3.8.14

**Given :**  $T_A = 26.7^\circ\text{C}$ ,  $T_B = 32^\circ\text{C}$ ,  
 $T_C = 36.93^\circ\text{C}$ ,  
 $L = 25 \text{ cm} = 0.25 \text{ m} = L_A = L_B = L_C$   
 $d = 10 \text{ mm} = 0.01 \text{ m}$ ,  
 $\theta_0 = T_0 - T_\infty = 100 - 20 = 80^\circ\text{C}$ ,  
 $\theta_L = T_L - T_\infty$   $h = 23 \text{ W/m}^2 \text{ K}$

Where,  $T_A, T_B, T_C$  are temperature at tip of the fins.

$\theta_0$  is temperature difference between base and ambient temperature.  $= T_0 - T_\infty$

$\theta_L$  is temperature difference between tip and ambient temperature.  $= T_L - T_\infty$

$$A = \text{cross sectional area of each rod} = \frac{\pi}{4} d^2$$

$$= \frac{\pi}{4} \times (0.01)^2 = 7.85 \times 10^{-5} \text{ m}^2$$

$$P = \text{Perimeter of each rod} = \pi d$$

$$= \pi \times 0.01 = 0.0314 \text{ m}$$

$$\theta_A = T_A - T_\infty = 26.7 - 20 = 6.7^\circ\text{C}$$

$$\theta_B = T_B - T_\infty = 32 - 20 = 12^\circ\text{C}$$

$$\theta_C = T_C - T_\infty = 36.93 - 20 = 16.93^\circ\text{C}$$

Considering bar as a infinitely long fin.

**(1) For rod A**

$$\frac{\theta_A}{\theta_0} = e^{-mx}$$

$$\frac{6.7}{80} = e^{-m \times 0.25} \quad [\text{as } x = L]$$

$$\therefore m = 9.92$$

$$m = \sqrt{\frac{h \cdot P}{k \cdot A}}$$

$$9.92 = \sqrt{\frac{23 \times 0.0314}{k_A \times 7.85 \times 10^{-5}}}$$

$$\therefore k_A = 93.49 \text{ W/mK} \quad \text{...Ans.}$$

**(2) For rod B**

$$\frac{\theta_B}{\theta_0} = e^{-mx}$$

$$\frac{12}{80} = e^{-m \times 0.25} \quad [\because x = L]$$

$\therefore$

$$m = 7.59$$

$$m = \sqrt{\frac{h \cdot P}{k \cdot A}}$$

$$7.59 = \sqrt{\frac{23 \times 0.0314}{k_B \times 7.85 \times 10^{-5}}}$$

$\therefore$

$$k_B = 159.70 \text{ W/mK} \quad \text{...Ans.}$$

**(3) For rod C**

$$\frac{\theta_C}{\theta_0} = e^{-mx}$$

$$\frac{16.93}{80} = e^{-m \times 0.25}$$

$$\therefore m = 6.21 = \sqrt{\frac{h \cdot P}{k_C \cdot A}} = \sqrt{\frac{23 \times 0.0314}{k_C \times 7.85 \times 10^{-5}}}$$

$$\therefore k_C = 238.43 \text{ W/mK} \quad \text{...Ans.}$$

**Ex. 3.8.15 :** A centrifugal pump which circulates hot liquid metal at  $500^\circ\text{C}$  is driven by a electricity motor, the motor is coupled to the pump impeller by a horizontal steel shaft 25 mm in diameter. The temperature of the motor is limited to a maximum value of  $60^\circ\text{C}$ , with ambient air at  $25^\circ\text{C}$ , what length of shaft should be specified between motor and the pump ? It may be presumed that the thermal conductivity of shaft material is  $35 \text{ W/mK}$  and convective film coefficient between steel shaft and air is  $157 \text{ W/m}^2 \text{ K}$ . Treat steel shaft as fin with insulated end.

**Soln. :** Refer P. 3.8.15

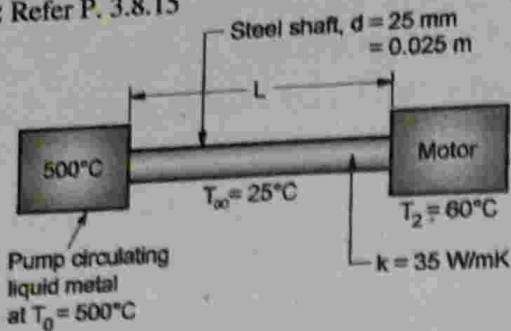


Fig. P. 3.8.15

**Given :**  $d = 25 \text{ mm} = 0.025 \text{ m}$ ,  
 $T_{\infty} = 25^{\circ}\text{C}$ ,  $k = 35 \text{ W/m}^{\circ}\text{K}$ ,  
 $T_0 = 500^{\circ}\text{C}$ ,  
 $T_L = 60^{\circ}\text{C}$ ,  $h = 157 \text{ W/m}^2 \text{ K}$

∴ One end of the shaft is connected to motor, therefore assuming no heat loss from fin tip.

The temperature distribution, treating the shaft as a fin insulated at the tip is given by,

$$\frac{\theta}{\theta_0} = \frac{T_L - T_{\infty}}{T_0 - T_{\infty}} = \frac{\cosh [m(L - x)]}{\cosh (mL)} \quad \dots \text{Equation (3.4.10)}$$

At  $x = L$ ,  $T = T_L$  the above Equation reduces to

$$\frac{T_L - T_{\infty}}{T_0 - T_{\infty}} = \frac{1}{\cosh (mL)}$$

$$\frac{60 - 25}{500 - 25} = \frac{1}{\cosh (mL)}$$

$$\therefore \cosh (mL) = 13.57$$

$$\therefore mL = 3.3 \quad \dots (i)$$

$$\text{But } m = \sqrt{\frac{hP}{kA}} = \sqrt{\frac{h \times \pi d}{k \times \frac{\pi}{4} d^2}} = \sqrt{\frac{4h}{kd}}$$

$$= \sqrt{\frac{4 \times 15.7}{35 \times 0.025}} = 8.472$$

$$\text{But } mL = 3.3$$

$$8.472L = 3.3$$

$$L = 0.3895 \text{ m} = 389.5 \text{ mm} \quad \dots \text{Ans.}$$

**Ex. 3.8.16 :** One end of a rod is inserted inside the furnace while the other end projects into surrounding air at  $30^{\circ}\text{C}$ . The diameter of the rod is  $1.5 \text{ cm}$ . Under steady state heat flow, the steady state temperatures at two points  $12 \text{ cm}$  apart on the rod were found to be  $120^{\circ}\text{C}$  and  $90^{\circ}\text{C}$  respectively. Assuming,  $h = 15 \text{ W/m}^2 \text{ K}$ , find the thermal conductivity of the rod material.

**Soln. :**  $d = 1.5 \text{ cm} = 0.015 \text{ m}$ ;  $T_{\infty} = 30^{\circ}\text{C}$

Distance between two points on rod,  $\delta x = 12 \text{ cm} = 0.12 \text{ m}$ ,  $h = 15 \text{ W/m}^2 \text{ K}$ ,

$T_1 = 120^{\circ}\text{C}$   
 and  $T_2 = 90^{\circ}\text{C}$

Refer Fig. P. 3.8.16. Considering infinitely long rod as fin.

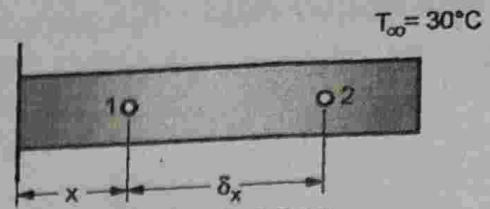


Fig. P. 3.8.16

Then, the temperature distribution along the rod is given as :

$$\frac{\theta}{\theta_0} = \frac{T - T_{\infty}}{T_0 - T_{\infty}} = e^{-mx} \quad \dots (a) \text{ (Equation 3.4.7)}$$

$$\theta_x = \theta_0 \cdot e^{-mx} \quad \dots (i)$$

Considering point 1 as base at temperature  $120^{\circ}\text{C}$  and point 2 at distance  $x$ , substituting in Equation (i)

$$T_x - T_{\infty} = (T_0 - T_{\infty}) e^{-mx}$$

$$90 - 30 = (120 - 30) e^{-mx \cdot 0.12}$$

$$\text{or } \frac{90}{60} = e^{m \times 0.12}$$

$$0.12 \times m = \ln \left( \frac{90}{60} \right) = 0.40545$$

$$\therefore m = 3.3789$$

$$m^2 = 11.4168$$

But

$$m^2 = \frac{h \cdot P}{k \cdot A}$$

$$m^2 = \frac{h \times \pi d}{k \times \frac{\pi}{4} \cdot d^2} = \frac{4h}{kd}$$

$$11.4168 = \frac{4 \times 15}{k \times 0.015}$$

$$k = 350.36 \text{ W/mK} \quad \dots \text{Ans.}$$

**Ex. 3.8.17 :** A metallic plate is maintained at  $250^{\circ}\text{C}$ . A  $2.5 \text{ cm}$  diameter aluminum rod protrudes  $10 \text{ cm}$  from this plate. Rod is exposed to atmosphere at  $20^{\circ}\text{C}$  with  $h = 15 \text{ W/m}^2 \text{ K}$ . Take  $K = 200 \text{ W/mK}$  for aluminum. Assuming the rod as a fin insulated at end, calculate heat loss by rod. Also calculate fin efficiency.

**Soln. :**

$$D = 0.025 \text{ m}$$

$$L = 0.01 \text{ m}$$

$$k = 200 \text{ W/mK}$$

$$T_0 = 250^{\circ}\text{C}$$

$$T_a = 20^{\circ}\text{C}$$

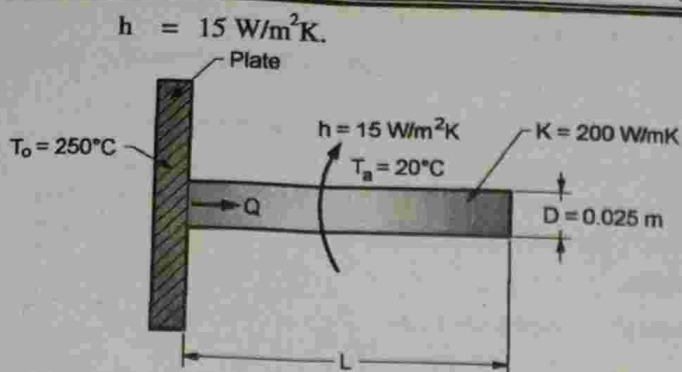


Fig. P. 3.8.17

Fin efficiency :

$$\eta = \frac{\tanh(m \cdot L)}{m \cdot L}$$

 $A_c$  is the area of cross section

$$A_c = \frac{\pi D^2}{4} \text{ m}^2 = \frac{\pi \times (0.025)^2}{4} = 4.9 \times 10^{-4} \text{ m}^2$$

$$P = \pi \cdot D = 3.14 \times 0.025 = 0.078 \text{ m}$$

$$\therefore m = \sqrt{\frac{h \cdot P}{k \cdot A_c}} = \sqrt{\frac{15 \times 0.078}{200 \times 4.9 \times 10^{-4}}} = 3.46 \text{ m}^{-1}$$

$$\eta = \frac{\tanh(m \cdot L)}{m \cdot L}$$

$$= 0.996 \text{ or } 99.6\%$$

...Ans.

**Ex. 3.8.18 :** Heat is being transferred from oil at  $90^\circ\text{C}$  to air at  $30^\circ\text{C}$  across a plane wall of area  $0.5 \text{ m} \times 0.5 \text{ m}$ . Heat transfer coefficient on oil side and air side are  $150 \text{ W/m}^2 \text{ K}$  and  $15 \text{ W/m}^2 \text{ K}$  respectively. It is desired to increase the total heat transfer rate by 250% by putting fins on air side. Calculate length of fins based on following data.

Thickness of fins = 2 mm,

Thermal conductivity =  $75 \text{ W/mK}$ 

Number of fins = 30 fins have insulated tip

Neglect wall thickness and assume  $5^\circ\text{C}$  drop in wall temperature after putting fins.**Soln. :**

$$T_{\text{oil}} = 90^\circ\text{C},$$

$$T_{\text{air}} = 30^\circ\text{C} = T_\infty$$

$$h_{\text{oil}} = 150 \text{ W/m}^2 \text{ K},$$

$$\text{thickness, } t = 2 \text{ mm} = 0.002 \text{ m}$$

$$h_{\text{air}} = 15 \text{ W/m}^2 \text{ K},$$

$$k_{\text{fin}} = 75 \text{ W/m K}$$

Number of fins on the surface are 30

$5^\circ\text{C}$  drop in wall temperature after putting fins, hence  $\Delta T = 5^\circ\text{C}$

$$\text{Area of wall surface, } A = 0.5 \times 0.5 = 0.25 \text{ m}^2$$

**Heat transfer from the plane wall where no fins fitted**

$$Q_1 = hA \Delta T = 15 \times 0.5 \times 0.5 (90 - 30) = 225 \text{ W}$$

(heat transfer rate shall be decided by lower  $h$  side)

It is now desired to increase this heat transfer rate by 250% by fitting fins on air side.

$$\text{Desired } Q = Q_1 + 250\% Q_1$$

$$= 225 + 2.5 \times 225 = 787.5 \text{ W}$$

$$\text{Area occupied by 30 fins} = 30 \times 0.002 \times 0.5 = 0.03 \text{ m}^2$$

$$\text{Unoccupied wall area} = 0.5 \times 0.5 - 0.03 = 0.22 \text{ m}^2$$

Heat transfer rate by unoccupied area of wall

$$= h_{\text{air}} \times A_{\text{un}} (T_{\text{oil}} - \Delta T - T_{\text{air}})$$

$$= 15 \times 0.22 (90 - 5 - 30) = 181.5 \text{ W}$$

Hence heat transfer desired by 30 fins

$$= 787.5 - 181.5 = 606 \text{ W}$$

$$\therefore \text{heat transfer rate per fin} = \frac{606}{30} = 20.2 \text{ W}$$

$$m = \sqrt{\frac{2h}{kt}} = \sqrt{\frac{2 \times 15}{75 \times 0.002}} = 14.142$$

$$Q_{\text{fin}} = kA_m \theta_b \tanh ml$$

$$20.2 = 75 \times 0.5 \times 0.002 \times 14.142$$

$$(90 - 5 - 30) \tanh (14.142 l)$$

$$\tanh (14.142 l) = 0.3463$$

$$14.142 l = 0.3612$$

$$l = 0.0255 \text{ m} = 2.55 \text{ cm} \text{ ...Ans.}$$

**Ex. 3.8.19 :** 10 rectangular fins of brass ( $k = 120 \text{ W/mK}$ ) are welded horizontally to a plane vertical surface of a tank, 1 m wide and 1 m high. The fins are 2 mm thick and 20 cm long. They are uniformly spaced on the vertical surface of tank, which is maintained at  $200^\circ\text{C}$ . If the unit is exposed to ambient air at  $20^\circ\text{C}$  with convective heat transfer coefficient of  $20 \text{ W/m}^2 \text{ K}$ , find.

(i) Heat transfer rate before putting the fins.

(ii) Heat transfer rate after putting the fins.

Prove that the fins are effective. Assume insulated end condition for the fins.

Soln. :

Given : Area of plate,  $A = 1 \text{ m} \times 1 \text{ m} = 1 \text{ m}^2$   
 Number of fins = 10,  $t = 2 \text{ mm} = 0.002 \text{ m}$   
 $L = 20 \text{ cm} = 0.2 \text{ m}$ ,  
 $k = 120 \text{ W/m K}$ ,  
 $T_{\text{wall}} = T_0 = 200^\circ\text{C}$   
 $T_\infty = 20^\circ\text{C}$ ,  
 $h = 20 \text{ W/m}^2 \text{ K}$

(i) Heat transfer rate before putting the fins

$$Q = h A (T_{\text{wall}} - T_a) \\ = 20 (1 \times 1) \times [200 - 20] = 3600 \text{ W}$$

(ii) Heat transfer rate after putting the fins

$$\text{Perimeter : } P = 2(b + t) = 2(1 + 0.002) = 2.004 \text{ m}$$

$$\text{Area : } A = b \times t = 1 \times 0.002 = 0.002 \text{ m}^2$$

$$m = \sqrt{\frac{hP}{kA}} = \sqrt{\frac{20 \times 2.004}{120 \times 0.002}} = 12.92$$

$$Q = \sqrt{PhkA} (T_0 - T_\infty) \tanh(mL)$$

$$\therefore Q = \sqrt{2.004 \times 20 \times 120 \times 0.002} \\ \times (200 - 20) \tanh(12.92 \times 0.2)$$

$$\therefore Q = 551.94 \text{ W/Pin}$$

Since there are 10 fins, therefore the total heat transfer,

$$Q_{\text{fins}} = 10 \times 551.94 = 5519.4 \text{ W} \quad \dots \text{Ans.}$$

Heat transfer from un finned ones

$$= h A_{\text{un}} \Delta T \\ = 20 \times (1 \times 1 - 10 \times 0.002) \times (200 - 20) \\ = 3528 \text{ W}$$

Total heat transfer of after fitting fins = 5519.4 + 3528

$$= 9.047.4 \text{ W} \quad \dots \text{Ans.}$$

(iii) Percentage increase in heat transfer after putting fins

$$= \frac{9047.4 - 3600}{3600} \times 100$$

$$= 151.3 \% \text{ rise in heat transfer rate.} \quad \dots \text{Ans.}$$

**Ex. 3.8.20 :** Fins, 12 in number, having  $k = 75 \text{ W/m K}$  and  $0.75 \text{ mm}$  thickness protrude  $25 \text{ mm}$  from a cylindrical surface of  $50 \text{ mm}$  diameter and  $1 \text{ m}$  length placed in an atmosphere of  $40^\circ\text{C}$ . If the cylindrical surface is maintained at  $150^\circ\text{C}$  and the heat transfer coefficient is  $23 \text{ W/m}^2 \text{ K}$ , calculate :

- the rate of heat transfer
  - the percentage increase in heat transfer due to fins
  - the temperature at the centre of fins and
  - the fin efficiency and the fin effectiveness.
- The fins are rectangular in shape and cross-section and they are attached along the length of the cylinder.

Soln. :

Given : Number of fins = 12,  $k = 75 \text{ W/mK}$ ,  
 $t = 0.75 \text{ mm} = 0.75 \times 10^{-3} \text{ m}$ ;  
 $L = 25 \text{ mm} = 0.025 \text{ m}$ ,

Cylinder diameter,

$$d = 50 \text{ mm} = 0.05 \text{ m},$$

$$W = 1 \text{ m}, T_\infty = 40^\circ\text{C},$$

$$T_0 = 150^\circ\text{C}, h = 23 \text{ W/m}^2 \text{ K}$$

Refer Fig. P. 3.8.20.

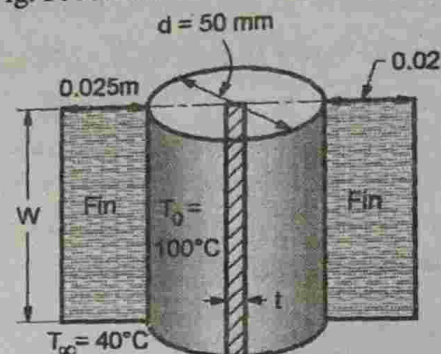


Fig. P. 3.8.20

In order to find heat transfer rate and apply appropriate Equation, we need to find whether fins are short or long or adequately long fins.

$$\text{Perimeter, } P = 2w = 2 \times 1 = 2 \text{ m}$$

$$A = W \times t = 1 \times (0.75 \times 10^{-3}) \\ = 0.75 \times 10^{-3} \text{ m}^2$$

$$m = \sqrt{\frac{h \cdot P}{k \cdot A}} = \sqrt{\frac{23 \times 2}{75 \times (0.75 \times 10^{-3})}} \\ = 28.284$$

$$m \cdot L = 28.284 \times 0.025 \\ = 0.707$$

Since  $mL$  is less than 3, it is the case a short fin.(a) Total heat transfer rate,  $Q_1$ 

$$Q_1 = k \cdot A \cdot m \cdot \theta_0 \times \left[ \frac{\tanh mL + \left(\frac{h}{mk}\right)}{1 + \frac{h}{mk} \cdot \tanh mL} \right] \dots (1)$$

$$\theta_0 = T_0 - T_\infty = 150 - 40 = 110^\circ\text{C}$$

$$\frac{h}{mk} = \frac{23}{28.284 \times 75} = 0.010843$$

On substituting the values in Equation (1),

$$\begin{aligned} Q_1 &= 75 \times (0.75 \times 10^{-3}) \times 28.284 \times 110 \\ &\quad \times \frac{\tanh(0.707) + 0.01083}{1 + 0.01083 \times \tanh(0.707)} \\ &= 107.73 \text{ W/fin} \end{aligned}$$

Hence total heat transfer for 12 fins,

$$\begin{aligned} Q_1 &= 12 \times Q_1 = 12 \times 107.73 \\ &= 1292.8 \text{ W} \end{aligned} \quad \dots \text{Ans.}$$

Bore or surface area of cylinder,

$$A_s = \pi d w = \pi \times 0.05 \times 1 = 0.1571 \text{ m}^2$$

$$\begin{aligned} \text{Finned area, } A_{\text{fin}} &= 12 \times A \\ &= 12 \times 0.75 \times 10^{-3} = 0.009 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Unfin area, } A_{\text{unfin}} &= A_s - A_{\text{fin}} = 0.1571 - 0.009 \\ &= 0.1481 \text{ m}^2 \end{aligned}$$

Heat transfer from unfin area,

$$\begin{aligned} Q_2 &= h \cdot A_{\text{unfin}} \cdot (T_o - T_\infty) \\ &= 23 \times 0.1481 \times (150 - 40) \\ &= 374.7 \end{aligned}$$

Total heat transfer rate,

$$\begin{aligned} Q_t &= Q_1 + Q_2 = 1292.8 + 374.7 \\ &= 1667.5 \text{ W} \end{aligned} \quad \dots \text{Ans.}$$

**(b) Percentage increase in heat transfer due to fins**

Heat transfer rate without fins on surface,

$$\begin{aligned} Q_s &= h A_s (T_o - T_\infty) = 23 \times 0.1571 \times 110 \\ &= 397.5 \text{ W} \end{aligned}$$

$$\begin{aligned} \text{Percentage increase} &= \left( \frac{Q_t - Q_s}{Q_s} \right) \times 100\% \\ &= \left( \frac{1667.5 - 397.5}{397.5} \right) \times 100 \\ &= 319.5\% \end{aligned} \quad \dots \text{Ans.}$$

**(c) Temperature at the centre of fins,**

$$T_x \text{ i.e. at } x = \frac{L}{2} = \frac{0.025}{2} = 0.0125$$

$$\frac{\theta}{\theta_0} = \frac{\cosh m(L-x) + \left(\frac{h}{mk}\right) \sinh m(L-x)}{\cosh mL + \left(\frac{h}{mk}\right) \sinh mL}$$

$$\frac{T_x - T_\infty}{T_o - T_\infty}$$

$$= \frac{\cosh(28.826 \times 0.0125) + 0.010843 \sinh(28.826 \times 0.0125)}{\cosh(0.707) + 0.010843 \sinh(0.707)}$$

$$\frac{T_x - 40}{150 - 40} = \frac{1.06954}{1.26883} = 0.8429; \quad T_x = 132.7^\circ\text{C}$$

$$\text{(d) Fin efficiency, } \eta_{\text{fin}} = \frac{Q_1}{h A_f \theta_0} \quad \because (A_f = P \cdot L)$$

$$\begin{aligned} \eta_{\text{fin}} &= \frac{107.73}{23 \times (2 \times 0.025) \times 110} \\ &= 0.8516 \text{ or } 85.16\% \end{aligned} \quad \dots \text{Ans.}$$

**(e) Effectiveness,**

$$E = \frac{Q_t}{Q_s} = \frac{1667.5}{397.5} = 4.195 \quad \dots \text{Ans.}$$

**Ex. 3.8.21 :** Blades of an aerofoil section are designed for a gas turbine to transfer 80 W of heat by each blade having the following dimensions :

$A_c = 2 \times 10^{-4} \text{ m}^2$ , Perimeter,

$P = 0.08 \text{ m}$ ,  $k = 25 \text{ W/mK}$ ,

Temperature of gas over the blades =  $800^\circ\text{C}$

Allowable temperature at the roots of the blade =  $350^\circ\text{C}$

Convective heat transfer coefficient  $h = 120 \text{ W/m}^2 \text{ K}$

Assuming blades as a fin with insulated ends, find the height of blade required.

**Soln. :**  $T_\infty = 800^\circ\text{C}$ ,  $T_0 = 350^\circ\text{C}$ ,

$Q = 80 \text{ W}$ ,  $\theta_0 = T_\infty - T_0 = 800 - 350 = 450^\circ\text{C}$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{120 \times 0.08}{25 \times (2 \times 10^{-4})}} = 43.82$$

$$Q = \sqrt{h \cdot P \cdot k \cdot A_c} \times \theta_0 \times \tanh(mL)$$

$$80 = \sqrt{120 \times 0.08 \times 25 \times 2 \times 10^{-4}} \times 450 \times \tanh(43.82 L)$$

$$\tanh(43.82 L) = 0.8114$$

$$43.82 L = 1.131$$

$$L = 0.0258$$

$$m = 2.58 \text{ cm} \quad \dots \text{Ans.}$$

**Ex. 3.8.22 :** If in Example 3.8.21 above, the blades are considered as finite long (Short) fin with convection heat transfer from its tips, find the length of blades required when all other data is remaining the same.

**Soln. :**

From Example 3.8.21 we have :

$$\theta_0 = 450^\circ\text{C}, m = 43.82;$$

$$\frac{h}{mk} = \frac{120}{43.82 \times 25} = 0.1095$$

Given : Finite long fin i.e. short fin and  $Q = Q_c$  using

Equation (3.4.17),

$$Q = \sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \left[ \frac{\tanh(mL) + \left(\frac{h}{mk}\right)}{1 + \left(\frac{h}{mk}\right) \tanh mL} \right]$$

$$80 = \sqrt{120 \times 0.08 \times 25 \times 2 \times 10^{-4}}$$

$$\times 450 \left[ \frac{\tanh(mL) + 0.1095}{1 + 0.1095 \tanh mL} \right]$$

$$0.8114 (1 + 0.1095 \tanh mL) = \tanh(mL) + 0.1095$$

$$\tanh(mL) = 0.7704$$

$$mL = 1.0213$$

$$43.82 \times L = 1.0213$$

$$L = 0.02331 \text{ m or } 2.331 \text{ cm} \quad \dots \text{Ans.}$$

**Ex. 3.8.23 :** A 1 m long, 10 cm diameter cylinder placed in atmosphere of  $35^\circ\text{C}$  is provided with 12 longitudinal fins ( $k = 100 \text{ W/m K}$ ), 1 mm thick. The fins protrude 3 cm from the cylinder surface. The heat transfer coefficient is  $25 \text{ W/m}^2 \text{ K}$ . Calculate the rate of heat transfer if the surface temperature of cylinder is at  $200^\circ\text{C}$ .

**Soln. :**

Given :  $d = 10 \text{ cm} = 0.1 \text{ m}$ ,  $W = 1 \text{ m}$ ,  $T_\infty = 35^\circ\text{C}$ ,

Number of fins,  $N = 12$ ,  $k = 100 \text{ W/mK}$

$t = 1 \text{ mm} = 0.001 \text{ m}$ ,  $L = 3 \text{ cm} = 0.03 \text{ m}$

$h = 25 \text{ W/m}^2 \text{ K}$ ,  $T_0 = 200^\circ\text{C}$

$$\text{Area per fin, } A = W \times t = 1 \times 0.001 = 0.001 \text{ m}^2$$

$$\text{Perimeter, } P = 2(W + t) = 2(1 + 0.001) = 2.002 \text{ m}$$

$$m = \sqrt{\frac{h \cdot P}{k \cdot A}} = \sqrt{\frac{25 \times 2.002}{100 \times 0.001}} = 22.372$$

$$m \cdot L = 22.372 \times 0.03 = 0.671$$

The cylinder has longitudinal fins. The fin is of finite length, it will dissipate heat by convection from its tip. Heat transfer rate from fins surface,

$$Q_{\text{fin}} = N \times \sqrt{h P k A} (T_0 - T_\infty) \tanh(mL) = 12 \times \sqrt{25 \times 2.002 \times 100 \times 0.001}$$

$$\begin{aligned} & \times (200 - 35) \tanh(0.67) \\ & = 2594.2 \text{ W} \\ Q_{\text{unfin area,}} & A_u = (\pi d W - N \cdot A) \\ & = (\pi \times 0.1 \times 1 - 12 \times 0.001) \\ & = 0.3022 \text{ m}^2 \\ Q_{\text{unfin}} & = h \cdot A_u (T_0 - T_\infty) \\ & = 25 \times 0.3022 \times (200 - 35) \\ & = 1246.4 \text{ W} \\ \therefore Q_{\text{total}} & = Q_{\text{fin}} + Q_{\text{unfin}} \\ & = 2594.2 + 1246.4 \\ & = 3840.6 \text{ W} \quad \dots \text{Ans.} \end{aligned}$$

**Ex. 3.8.24 :** An electric motor 300 mm long dissipating heat at a rate of 350 W is required to be fitted with plate fins radially outwards so that motor surface temperature does not exceed  $50^\circ\text{C}$ . Plate fins are 15 mm thick of 50 mm length (height from motor surface) with  $k = 40 \text{ W/m}^\circ\text{C}$ . Motor is exposed to atmosphere at  $25^\circ\text{C}$  with convective heat transfer coefficient of  $20 \text{ W/m}^2 \text{ K}$ . Determine number of fins required neglecting convection from tip of fins. Ignore heat convection from unfinned area of motor's outer surface.

**Soln. :**

Given :  $Q_{\text{motor}} = 350 \text{ W}$

$k = 40 \text{ W/m}^\circ\text{C}$

$T_{\text{base}} = 50^\circ\text{C}$ ;  $T_a = 25^\circ\text{C}$

$h = 20 \text{ W/m}^2 \text{ K}$

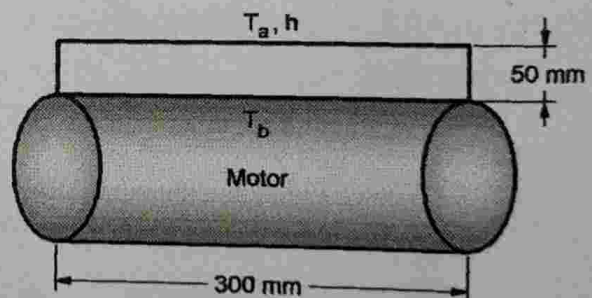


Fig. P. 3.8.24

Since convection from tip of fins is to be neglected, these are adequately long fin.

Heat dissipation from adequately long fin is given as ;

$$Q = m A k \theta_b \tanh mL$$

$$m = \sqrt{\frac{hP}{kA}}$$

$$= \sqrt{\frac{20 \times (0.3 + 0.3 + 0.015 + 0.015)}{40 \times 0.3 \times 0.015}}$$

$$= 8.367$$

$$\therefore Q = 8.367 \times 0.3 \times 0.015 \times 40 \times (50 - 25) \tanh(8.367 \times 0.05)$$

$$= 37.65 \tanh(0.418) = 14.89 \text{ W/fin}$$

Hence number of fins required to dissipate 350 W,

$$= \frac{350}{14.89} = 23.5 \Rightarrow 24 \text{ fins} \quad \dots \text{Ans.}$$

**Ex. 3.8.25 :** A frying pan with a long handle of rectangular cross section is used to heat the oil for frying purpose upto  $150^\circ\text{C}$ . Section of handle is  $5 \text{ mm} \times 10 \text{ mm}$ . Find the length of handle from the base of frying pan such that its temperature should not exceed  $30^\circ\text{C}$ . Assume,  $k = 260 \text{ W/m K}$ , Surrounding temperature  $20^\circ\text{C}$  and convectivity,  $h = 30 \text{ W/m}^2 \text{ K}$ .

**Soln. :** Refer Fig. P. 3.8.25

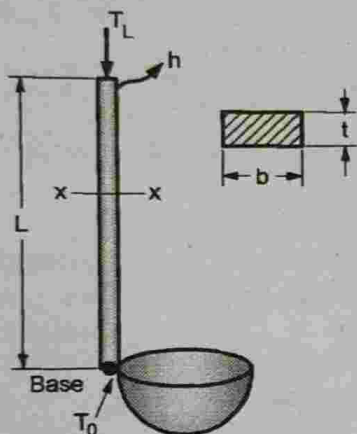


Fig. P. 3.8.25

$$T_0 = 150^\circ\text{C}, \quad b = 10 \text{ mm} = 0.01 \text{ m}$$

$$t = 5 \text{ mm} = 0.005 \text{ m}, \quad T_\infty = 20^\circ\text{C},$$

$$k = 260 \text{ W/m K}, \quad h = 30 \text{ W/m}^2 \text{ K},$$

$$T_L = 30^\circ\text{C}, \quad L = ?$$

$$P = 2(b + t) = 2(0.01 + 0.005) = 0.03 \text{ m}$$

$$A = b \times t = 0.01 \times 0.005 = 5 \times 10^{-5} \text{ m}^2$$

The long handle can be treated as an infinitely long fin. The temperature distribution along the handle from its base is given as :

$$\frac{T_L - T_\infty}{T_0 - T_\infty} = e^{-mx}$$

$$\frac{30 - 20}{150 - 20} = e^{-mL}$$

$$e^{-mL} = 0.07692$$

$$-mL = \ln(0.07692) = -2.565$$

$$mL = 2.565 \quad \dots (i)$$

$$\text{But, } m = \sqrt{\frac{h \cdot P}{k \cdot A}} = \sqrt{\frac{30 \times 0.03}{260 \times 5 \times 10^{-5}}} = 8.32$$

$$\therefore L = \frac{2.565}{m} = \frac{2.565}{8.32}$$

$$= 0.3083 \text{ m or } 30.83 \text{ cm} \quad \dots \text{Ans.}$$

**Ex. 3.8.26 :** In an experiment to determine the thermal conductivity of a very long solid rod of  $2.5 \text{ cm}$  diameter, its one end is placed in a furnace and rod is rejecting into a room with ambient air at  $22^\circ\text{C}$ . After steady-state conditions are achieved, the temperatures at two points  $10 \text{ cm}$  apart, are found to be  $110^\circ\text{C}$  and  $85^\circ\text{C}$  respectively. Convective heat transfer coefficient between the rod surface and the surrounding air is  $28.4 \text{ W/m}^2 \text{ K}$ . Determine the thermal conductivity of the rod material.

**Soln. :**

$$\text{Given : } d = 2.5 \text{ cm}, \quad T_\infty = 22^\circ\text{C},$$

$$T_b = 110^\circ\text{C}, \quad T = 85^\circ\text{C},$$

$$h = 28.4 \text{ W/m}^2 \text{ K}$$

Since temperatures are given, temperature expression is required to be used. As it is a case of very long fin, temperature expression will be,

$$\frac{\theta}{\theta_b} = e^{-mx}$$

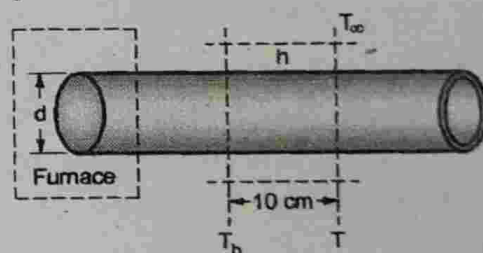


Fig. P. 3.8.26

$$\text{BC(1) } \quad \text{At } x = 0 \text{ (First temp point); } T = T_b = 110^\circ\text{C}$$

$$\text{BC(2) } \quad \text{At } x = 0.1 \text{ m; } T = 85^\circ\text{C}$$

Substituting,

$$\frac{85 - 22}{110 - 22} = e^{-0.1m}$$

$$\Rightarrow m = 3.34 = \sqrt{\frac{hP}{kA}}$$



$$= \sqrt{\frac{28.4 \times \pi \times 0.025}{k \frac{\pi}{4} \times 0.025^2}}$$

$$\therefore k = 406 \text{ W/mK}$$

...Ans.

**Ex. 3.8.27 :** A heating unit is made in the form of a vertical tube fitted with rectangular section steel fins. The tube height is 1.2 m and its outer diameter is 60 mm. The fins are 50 mm in height and their thickness is 3 mm. The total number of fins used is 20. The temperature of the base of the fin is 80°C and surrounding temperature is 18°C. The HTC on the fin surface and tube surface to the surrounding air is 9.3 W/m<sup>2</sup>K.  $k$  (fin material) = 55.7 W/m°C. Calculate the rate of heat transfer from the tube with and w/o fin.

**Soln. :**

**Given :**  $d = 60 \text{ mm}$ ,  $l = 50 \text{ mm}$ ,  
 $t = 3 \text{ mm}$ ,  $n = 20$ ,  
 $T_b = 80^\circ \text{C}$ ,  $T_\infty = 18^\circ \text{C}$ ,  
 $h = 9.3 \text{ W/m}^2 \text{K}$ ,  
 $k = 55.7 \text{ W/m}^\circ \text{C}$

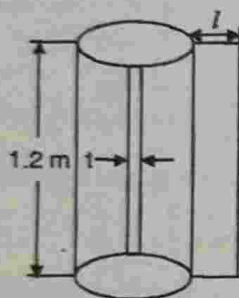


Fig. P. 3.8.27

We must decide type of fins.

$$m = \sqrt{\frac{Ph}{kA}} = \sqrt{\frac{2 \times 1.2 \times 9.3}{55.7 \times 1.2 \times 0.003}}$$

$$= 10.55$$

$$ml = 10.55 \times 0.05$$

$= 0.527 \ll 3$  hence fins to be treated as short fin

Heat flow rate from a short fin,

$$Q = mAk\theta_b \frac{\frac{h}{mk} + \tan hml}{1 + \frac{h}{mk} + \tan ml}$$

$$\frac{h}{mk} = \frac{9.3}{10.5 \times 55.7} = 0.0159$$

$$Q = 10.55 \times 1.2 \times 0.003 \times 55.7$$

$$\times (80 - 18) \times \frac{0.0159 + \tan h(0.527)}{1 + 0.0159 \tan h(0.527)}$$

$$= 131.16 \times 0.499 = 65.45 \text{ W}$$

$$\text{Heat transfer from 20 fins} = 20 \times 65.45$$

$$= 1309 \text{ W}$$

$$\text{Area of tube} = \pi DL = \pi \times 0.06 \times 1.2$$

$$= 0.2261 \text{ m}^2$$

$$\text{Area occupied by 20 fins} = 20 \times 1.2 \times 0.003$$

$$= 0.072 \text{ m}^2$$

$$\text{Unfinned area of the tube} = 0.2261 - 0.072$$

$$= 0.1541 \text{ m}^2$$

Heat flow from unfinned area of tube

$$= hA\theta_b = 9.3 \times 0.1541 \times (80 - 18)$$

$$= 88.85 \text{ W}$$

Total heat flow from finned tube = 1309 + 88.85

$$= 1397.85 \text{ W}$$

...Ans.

Heat flow from tube w/o fins,

$$Q = hA\theta_b = 9.3 \times 0.2261 \times (80 - 18)$$

$$= 130.37 \text{ W}$$

...Ans.

**Ex. 3.8.28 :** The end of a very long cylindrical stainless steel rod is attached to a heated wall and its surface is in contact with a cold fluid. Determine by what percentage the heat removal rate would change.

(i) If the rod diameter is doubled.

(ii) If the rod is made up of aluminium.

Use  $k$  for Al = 204.7 W/mK.  $k$  for S.S. = 16.17 W/mK.

**Soln. :** For fin of infinite length, heat transferred  $Q$ ,

$$Q = \sqrt{k \cdot A \cdot h \cdot P} (T_0 - T_\infty) \quad \dots(1)$$

(i) Percentage change in heat removal rate if rod diameter is doubled, i.e.

$$d_2 = 2d_1,$$

Cross sectional area,

$$A_1 = \frac{\pi}{4} d_1^2; \quad A_2 = \frac{\pi}{4} (2d_1)^2 = 4 \left( \frac{\pi}{4} d_1^2 \right) = 4A_1$$

Perimeter,

$$P_1 = \pi d_1; \quad P_2 = \pi d_2 = \pi \times 2d_1 = 2P_1$$

Using Equation (1), we can write,

$$\frac{Q_1}{Q_2} = \frac{\sqrt{h \cdot P_1 \cdot k \cdot A_1} (T_0 - T_\infty)}{\sqrt{h \cdot P_2 \cdot k \cdot A_2} (T_0 - T_\infty)}$$

$$\frac{Q_1}{Q_2} = \sqrt{\frac{P_1 \cdot A_1}{P_2 \cdot A_2}} = \sqrt{\frac{P_1}{2P_1} \times \frac{A_1}{4A_1}} = \sqrt{\frac{1}{8}}$$

$$Q_2 = 2\sqrt{2} Q_1$$

$$\therefore \% \text{ change in heat removal rate} = \frac{Q_2 - Q_1}{Q_1} \times 100 \%$$

$$= \frac{(2\sqrt{2} \cdot Q_1 - Q_1)}{Q_1} \times 100$$

$$= 182.84 \% \text{ (Increase)} \quad \dots \text{Ans.}$$

(ii) Percentage change in heat removal rate if the stainless steel rod is replaced by aluminium rod.

For stainless steel fin,  $k = k_1 = 16.17 \text{ W/mK}$

$$\therefore Q_1 = \sqrt{h P k_1 A} (T_0 - T_\infty)$$

$$= \sqrt{16.17} \sqrt{h P A} (T_0 - T_\infty)$$

$$Q_1 = 4.0212 \sqrt{h P A} (T_0 - T_\infty)$$

For Aluminum fin,  $k_{Al} = k_2 = 204.7 \text{ W/mK}$

$$Q_2 = \sqrt{h P k_2 A} (T_0 - T_\infty)$$

$$= \sqrt{204.7} \sqrt{h P A} (T_0 - T_\infty)$$

$$Q_2 = 14.307 \sqrt{h P A} (T_0 - T_\infty)$$

Percentage change in heat removal rate

$$= \left( \frac{Q_2 - Q_1}{Q_1} \right) \times 100 \%$$

$$= \frac{(14.307 - 4.0212)}{4.0212} \times 100$$

$$= 255.798 \% \text{ (Increase)} \quad \dots \text{Ans.}$$

**Ex. 3.8.29 :** Three 10 mm diameter rods A, B and C protrude from a steam bath at  $100^\circ\text{C}$  to a length of 25 cm into atmosphere at  $20^\circ\text{C}$ . The temperature at the other ends are found to be  $26.76^\circ\text{C}$  for "A",  $32.0^\circ\text{C}$  for "B" and  $36.93^\circ\text{C}$  for "C". Neglecting the effect of radiation and assuming the surface film coefficient of heat transfer as  $23 \text{ W/m}^2\text{K}$ , evaluate their thermal conductivities. Assume insulated tip condition for fin analysis.

**Soln. :**

**Given :**  $D = 10 \text{ mm}$ ,  $h = 23 \text{ W/m}^2\text{K}$   
 $l = 25 \text{ cm}$ ,  $T_\infty = 20^\circ\text{C}$   
 $T_b = 100^\circ\text{C}$ ,  $T_{eA} = 26.76^\circ\text{C}$   
 $T_{eB} = 32^\circ\text{C}$ ,  $T_{eC} = 36.93^\circ\text{C}$

Fins are insulated tips

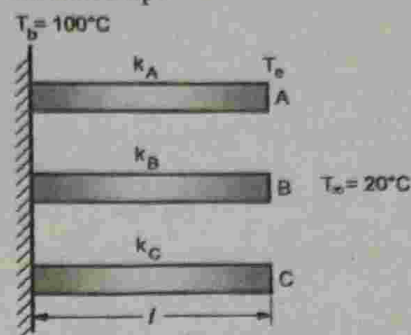


Fig. P. 3.8.29

**Rod 'A' :** Temperature at the end of fin,

$$\frac{\theta_e}{\theta_b} = \frac{T_{eA} - T_\infty}{T_b - T_\infty} = \frac{1}{\cosh ml}$$

$$m = \sqrt{\frac{Ph}{k_A \cdot A}} = \sqrt{\frac{\pi D \times h}{k_A \times \frac{\pi}{4} D^2}}$$

$$= \sqrt{\frac{4h}{k_A D}}$$

$$= \sqrt{\frac{4 \times 23}{k_A \times 0.01}} = \frac{95.9}{\sqrt{k_A}}$$

Substituting values

$$\frac{26.76 - 20}{100 - 20} = \frac{1}{\cosh \left( \frac{95.9 \times 0.25}{\sqrt{k_A}} \right)}$$

$$\cosh \left( \frac{23.98}{\sqrt{k_A}} \right) = 11.83$$

$$\frac{23.98}{\sqrt{k_A}} = \cosh(11.8) = 3.16$$

$$k_A = 57.6 \text{ W/mK} \quad \dots \text{Ans.}$$

**Rod 'B' :**

$$m = \frac{95.9}{\sqrt{k_B}}$$

$$\frac{T_{eB} - T_\infty}{T_b - T_\infty} = \frac{1}{\cosh \left( \frac{95.9 \times 0.25}{\sqrt{k_B}} \right)}$$

$$\frac{32 - 20}{100 - 20} = \frac{1}{\cosh \left( \frac{23.98}{\sqrt{k_B}} \right)}$$

$$\cosh \left( \frac{23.98}{\sqrt{k_B}} \right) = 6.667$$

$$\text{or } \frac{23.98}{\sqrt{k_B}} = 2.58$$

$$k_B = 86 \text{ W/mK} \quad \dots \text{Ans.}$$



Rod 'C' :

$$m = \frac{95.9}{\sqrt{k_c}}$$

$$\frac{T_{ec} - T_\infty}{T_b - T_\infty} = \frac{1}{\cosh\left(\frac{95.9 \times 0.25}{\sqrt{k_c}}\right)}$$

$$\frac{36.93 - 20}{100 - 20} = \frac{1}{\cosh\left(\frac{23.98}{\sqrt{k_c}}\right)}$$

$$\cosh\left(\frac{23.98}{\sqrt{k_c}}\right) = 4.725$$

$$\frac{23.98}{\sqrt{k_c}} = 2.235$$

$$k_c = 115 \text{ W/mK} \quad \dots \text{Ans.}$$

**Summary**

- Fins or extended surfaces are used to increase the heat transfer rates by increasing the surface area of solid surfaces so that the temperatures can be maintained within specified limits.

- Temperature distribution and Heat transfer in case of fins are :

| Case | Condition                                              | $\frac{\theta}{\theta_0} = \frac{T_L - T_\infty}{T_0 - T_\infty}$                                                                       | Heat transfer, Q                                                                                                                                     |
|------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| I    | Infinitely long fin.                                   | $e^{-mx}$                                                                                                                               | $\sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0$                                                                                                    |
| II   | Adequately long fin with insulated tip                 | $\frac{\theta}{\theta_0} = \frac{\cosh m(L-x)}{\cosh mL}$                                                                               | $\sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \cdot \tanh mL$                                                                                     |
| III  | Finite / short fin.<br>Convection heat transfer at tip | $\frac{\theta}{\theta_0} = \frac{\cosh m(L-x) + \left(\frac{h}{mk}\right) \sinh m(L-x)}{\cosh mL + \left(\frac{h}{mk}\right) \sinh mL}$ | $\sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \left[ \frac{\tanh mL + \left(\frac{h}{mk}\right)}{1 + \left(\frac{h}{mk}\right) \tanh mL} \right]$ |

- If  $\tanh mL \geq 0.99$ , it is considered an infinite long fin.
- Effectiveness of fin, E :

$$E = \frac{\text{Actual heat transfer from finned surface, } Q}{\text{Heat transfer without fins}}$$

For fins with insulated tip :

$$E = \sqrt{\frac{k \cdot P}{h \cdot A}} \tanh mL$$

- Fin efficiency,  $\eta = \frac{Q}{Q_{\max}} = \frac{Q}{h \cdot P \cdot L \cdot \theta_0}$  (Where,  $P \cdot L = A_f$ )

- Types of fins are rectangular, tapered, radial plate fins, annular fins and pin fins.

The fins attached to plain walls are called **straight fins** and the fins attached in circumferential direction to a cylinder are called **annular fins**.

- Geometry of fins are selected based on cost, efficiency, performance, weight, availability of space and pressure drop in the heat transfer fluid etc.
- Assumption in analysis of rectangular or pin fins are : heat conduction is under steady state in one direction, coefficient of convective and conductive heat transfers are constant. No heat generation within fin, material is isotropic and wall / base temperature is maintained constant.

General solution is :

$$\theta = C_1 e^{mx} + C_2 e^{-mx} \quad \left( \text{where, } m = \sqrt{\frac{h \cdot P}{k \cdot A}} \right)$$

- Overall fin effectiveness,  $E_0$  :

$$E_0 = \frac{Q_{\text{total}} \text{ (from finned surface)}}{Q_{\text{nofins}} \text{ (from base surface if there are no fins)}}$$

$$= \frac{h \cdot (A_{\text{unfin}} + \eta \cdot A_{\text{fin}}) (T_0 - T_{\infty})}{h \cdot A_{\text{nofin}} (T_0 - T_{\infty})}$$

- Area weighted fin efficiency,  $\eta_{\text{total}}$  :

$$\eta_{\text{total}} = \frac{Q_{\text{total}} \text{ from finned surface}}{Q \text{ if total area were maintained at the base temperature } T_0}$$

$$= 1 - \frac{A_{\text{fin}}}{A_{\text{total}}} (1 - \eta)$$

### Exercise

[Note : For answers, please refer the section number indicated in bracket.]

- Q. 1 Why fins are used ? State some of its applications. [Sections 3.1 and 3.2]
- Q. 2 Explain straight and annular fins. [Section 3.2]
- Q. 3 Show that the heat transfer rate in case of infinite long fin is given as :
- $$Q = \sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \quad [\text{Section 3.3.2}]$$
- Q. 4 State and derive the condition for an infinitely long fin. [Section 3.3.1]
- Q. 5 Why the fins are provided (on gas side) having low thermal conductivity. [Section 3.1]
- Q. 6 Derive an expression for heat transfer for an adequately long fin with insulated tip. [Section 3.3.2]
- (Ans. :  $Q = \sqrt{h \cdot P \cdot k \cdot A} \cdot \theta_0 \tanh mL$ )

- Q. 7 What do you understand by short fins ? Show that the temperature distribution in such a long fin is given as :

$$\frac{\theta}{\theta_0} = \frac{\cosh m(L-x) + \left(\frac{h}{mk}\right) \sinh m(L-x)}{\cosh mL + \left(\frac{h}{mk}\right) \sinh mL}$$

### [Section 3.3.3]

- Q. 8 Define effectiveness of a fin. Does the effectiveness of a fin always increases ? Justify ? [Section 3.4]
- Q. 9 How the effectiveness of fins can be increased ? [Section 3.4]
- Q. 10 Define fin efficiency. Prove that :
- $$\eta_{\text{fin}} = \frac{\tanh mL}{mL} \text{ for long fin with insulated tip}$$
- [Section 3.5]
- Q. 11 What do you understand by overall fin effectiveness ? Show that the total efficiency of the fin is :
- $$\eta_{\text{total}} = 1 - \frac{A_{\text{fin}}}{A_{\text{total}}} (1 - \eta)$$
- [Sections 3.6 and 3.6.1]
- Q. 12 Why error occurs in temperature measurement of gases by thermometer ? Explain with the help of a neat sketch. [Section 3.7]
- Q. 13 Show that the error in temperature measurement ( $T_{\infty} - T_L$ ) equals to  $\sqrt{\frac{h}{k \cdot t}}$ .
- [Section 3.7]